

Appendix for: How accurate is genomic prediction across wild
populations?

S1: Genomic animal models

To estimate variance components and heritabilities in subpopulations, we fitted 9 genomic animal models, one for each trait in each scenario. These models were of the same form as the GP models described in the main text, except all available data for a given trait/scenario was used to train the models (*i.e.*, there was no test set). The same fixed and random effects as in Equation (1) in the main text were included, the GRM was constructed with the same procedure, and the same priors were used in R-INLA.

By sampling from the posteriors of these respective animal models, we estimated heritabilities and variance components in subsets of the data. Additive genetic variances and heritabilities were estimated as the posterior means

$$\hat{\sigma}_G^2 = \frac{1}{S} \sum_{s=1}^S \hat{\sigma}_{G,(s)}^2 = \frac{1}{S} \sum_{s=1}^S \widehat{\text{Var}}\left(g_i^{(s)}\right), \quad \hat{h}_{AM}^2 = \frac{1}{S} \sum_{s=1}^S h_{(s)}^2 = \frac{1}{S} \sum_{s=1}^S \frac{\widehat{\text{Var}}\left(g_i^{(s)}\right)}{\widehat{\text{Var}}\left(\tilde{y}_{ij}^{(s)}\right)}, \quad (\text{S1})$$

respectively, where $\hat{\sigma}_G^2$ is the estimated additive genetic variance, $S = 10\,000$ is the chosen number of posterior samples, the index (s) indicates the s^{th} posterior sample, $\hat{\sigma}_{G,(s)}^2$ is a posterior sample of the additive genetic variance, $g_i^{(s)}$ is a posterior sample of the genetic value, $\hat{h}_{AM}^2$ is the estimated heritability, $h_{(s)}^2$ is a posterior sample of the heritability, $\tilde{y}_{ij}^{(s)}$ is a posterior sample of the linear predictor from the model excluding ε_{ijk} (see Equation (1) in the main text) and $\widehat{\text{Var}}$ denotes sample variance. We use $\tilde{y}_{ij}$ (rather than the data y_{ijk}) to exclude the variance caused by the measurement error ε_{ijk} from the denominator, since it is not a biologically relevant source of variation (see de Villemereuil *et al.*, 2018). This is in contrast to the accuracy scaling factor λ^2 used in the main text, which *does* use the sample variance in the data y_{ijk} in its denominator.

Here, $\widehat{\text{Var}}$ denotes sample variance taken over the individuals i in a given subset of the individuals in a given scenario (Sorensen *et al.*, 2001; Lara *et al.*, 2022). This lets us find additive genetic variances and heritabilities in subsets of the data, for example in the test set of a given GP model. We use the estimates $\hat{\sigma}_G^2$ of additive genetic variance in a given test set to find the proper scaling factor λ for GP accuracy (see main text and Appendix S2), and the estimates $\hat{h}_{AM}^2$ are used in Appendix S3 in the context of formulas for expected accuracy, and are also used in Appendix S4 in comparisons the test set heritabilities derived from the GP models in the main text. Results for the posterior statistics for the genomic animal are shown in Appendix S6.

S2: GP accuracy metrics

Derivation of accuracy scaling factor

Here we show that our accuracy measure (Equation (3) in the main text) is equivalent to $r = \text{Corr}(g_i, \hat{g}_i)$. We assume that among all the model terms in y_{ijk} , the only one that covaries with $\hat{g}_i$ is g_i , which essentially means that our model is correctly specified, and so we have accounted for all confounding factors. Thus, we can write

$$\begin{aligned}
 \text{Corr}(y_{ijk}, \hat{g}_i) &= \frac{\text{Cov}(y_{ijk}, \hat{g}_i)}{\sqrt{\text{Var}(y_{ijk}) \text{Var}(\hat{g}_i)}} \\
 &= \frac{\text{Cov}(\dots + g_i + \dots, \hat{g}_i)}{\sqrt{\text{Var}(y_{ijk}) \text{Var}(\hat{g}_i)}} \\
 &= \frac{\text{Cov}(g_i, \hat{g}_i)}{\sqrt{\text{Var}(y_{ijk}) \text{Var}(\hat{g}_i)}} \\
 &= \frac{\text{Cov}(g_i, \hat{g}_i)}{\sqrt{\text{Var}(y_{ijk}) \text{Var}(\hat{g}_i)}} \cdot \sqrt{\frac{\text{Var}(g_i)}{\text{Var}(g_i)}} \\
 &= \frac{\text{Cov}(g_i, \hat{g}_i)}{\sqrt{\text{Var}(g_i) \text{Var}(\hat{g}_i)}} \cdot \sqrt{\frac{\text{Var}(g_i)}{\text{Var}(y_{ijk})}} \\
 &= \text{Corr}(g_i, \hat{g}_i) \cdot \lambda, \\
 &= r \cdot \lambda,
 \end{aligned}$$

where $\lambda = \sqrt{\frac{\text{Var}(g_i)}{\text{Var}(y_{ijk})}}$ as defined in the main text. In other words, we define λ^2 as the variance ratio between the genetic values and the phenotypic measurements, *including the measurement error*. As discussed by Wilson (2008) and de Villemereuil *et al.* (2018), the proper definition of heritability should not involve non-biological sources of variation such as measurement error in the denominator. Thus, we should not write $\lambda = h$ as is commonly done in the literature, as λ will only be equal to the square root of the heritability under a naïve definition of heritability.

Uncertainty in GP accuracy

Assessing the uncertainty in the GP accuracy $\hat{r}$ is straightforward in our within-population models. We can use $\hat{\sigma}_{\text{CV}}$ to estimate the uncertainty, where $\hat{\sigma}_{\text{CV}}$ is the sample standard deviation over the 10 accuracies obtained from a given 10-fold cross-validation. However, for the across-population GP models, we only get a single point estimate of accuracy per model. One might think that it would have been enough to take joint posterior samples of the genetic value vector $\mathbf{g}$ from a given across-population GP model, and use these each of these samples in Equation (3) in the main text, but doing this will severely underestimate the accuracy compared to using $\hat{g}_i$. The joint posterior of the GP models give

us access to the uncertainty in g_i , but what we are interested in is the uncertainty in the point estimate $\hat{g}_i$. Thus, we need multiple samples of $\hat{g}_i$, which implies a need for refitting the model.

Here we suggest a procedure that involves resampling from the joint posterior of the fitted models, which is similar to a parametric bootstrap. For each GP model, we drew 100 response vector samples $\mathbf{y}^{(b)}$ ($b = 1, \dots, 100$) from the joint posterior distribution of the response vector entries. For each new resampled response vector $\mathbf{y}^{(b)}$ we then refitted the model and retained the accuracy of the genetic values obtained from the refitted model, $\hat{r}^{(b)}$, calculated using Equation (3) with g_i and y_{ijk} replaced by $g_i^{(b)}$ and $y_{ijk}^{(b)}$, respectively. This procedure thus produces 100 replicates $\hat{r}^{(b)}$ of the accuracy $\hat{r}$ of a given GP model, and we can take the sample standard deviation of these replicates to estimate the uncertainty of $\hat{r}$, or look at other summary statistics such as quantiles.

The resampling and refitting procedure has the obvious downside of being computationally cumbersome, as it requires fitting each GP a large number of additional times, *e.g.*, 100 times. Because of the large computational requirements, we only estimated the uncertainty in a subset of the across-population GP models, namely the models for tarsus length in Scenario 1 (Figure S1), as well as the all models in Scenario 2 (Figure S2).

In Figure S3 we demonstrate that the uncertainties from the resampling method are well-calibrated. That is, for a given probability level $\pi \in (0, 1)$, we compute the empirical coverage as the proportion of models for which the realized accuracy $\hat{r}$ falls within the central π -quantile range $[0.5 - \frac{\pi}{2}, 0.5 + \frac{\pi}{2}]$ of the corresponding resampled accuracies $\hat{r}^{(b)}$. We find a close correspondence between this empirical proportion and its theoretical expectation (which is equal to π), indicating a well-calibrated uncertainty measure.

S3: Analysis of formulas for expected accuracy

Here we investigate the efficacy of the expected accuracy formula

$$E(r) \approx \sqrt{\frac{1}{1 + \frac{M_e}{Nh^2}}}, \quad (S2)$$

given as equation (4) in the main text, and the more general version

$$E(r)^2 = \frac{\frac{h_M^2}{h^2}}{1 + \frac{M_e}{N \cdot h_M^2} \cdot (1 - h^2 E(r)^2)}, \quad (S3)$$

given by Wray *et al.* (2013). Here $E(r)$, h^2 , N and M_e are as given in the main text, while h_M^2 is the heritability captured by M genetic markers. By differentiating between h^2 and h_M^2 , the generalized formula (S3) allows for the possibility that one has not included enough markers to capture all the

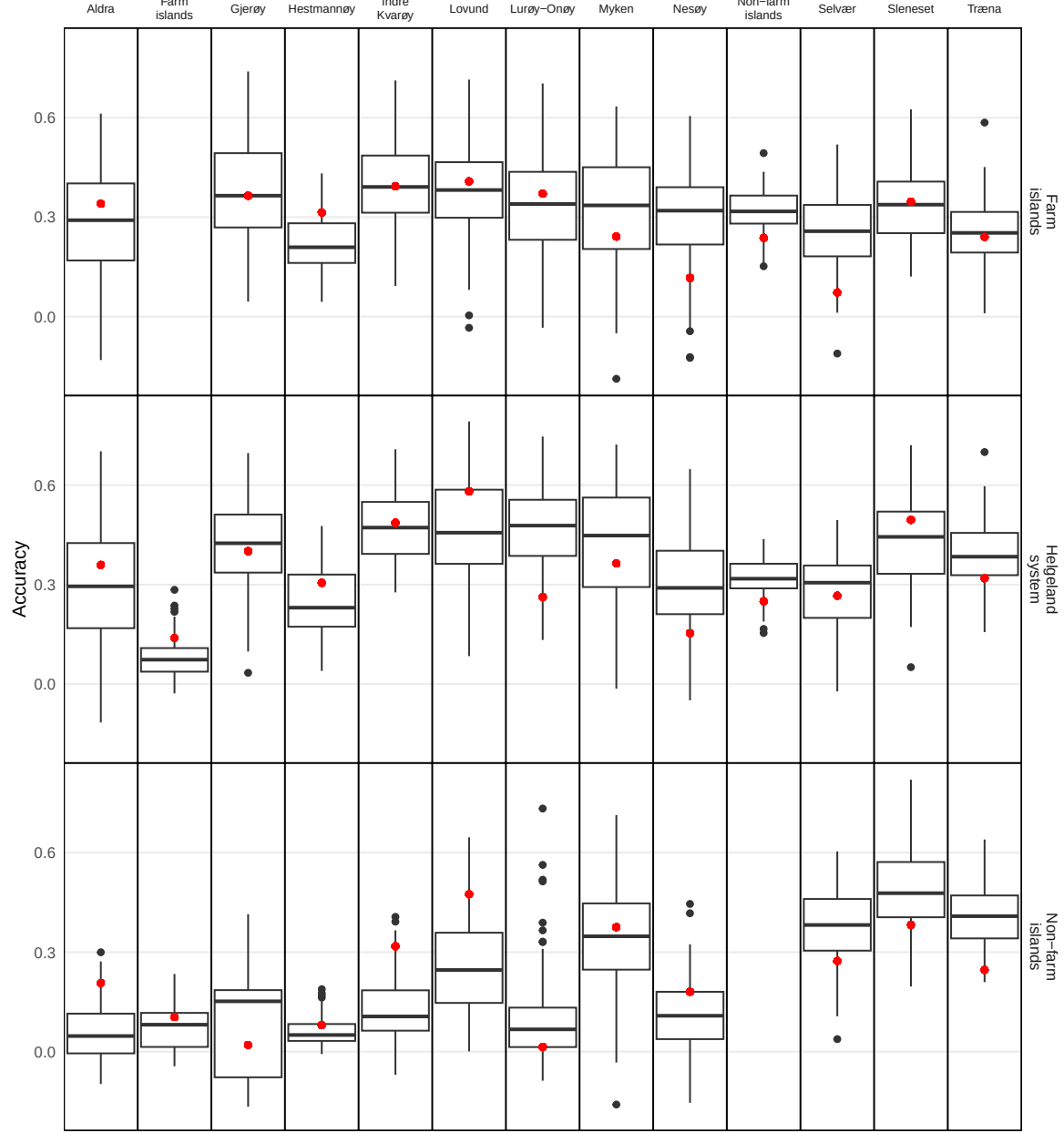


Figure S1: Accuracy in the across-population GP models for tarsus length. Columns correspond to test sets, and rows correspond to training sets, excluding the test island(s) if relevant. The red dots are the point estimates of accuracy obtained from the original GP model and presented in the main text, while the box plots are made from the accuracies from the refitted models.

additive genetic effects, that is, to tag all the QTL of the polygenic trait. And, in contrast to formula (S2), the generalization (S3) contains the factor $(1 - h^2 E(r)^2)$ in the denominator, which makes the equation quadratic in $E(r)^2$. By assuming $h^2 = h_M^2$ (*i.e.*, *marker saturation*, the markers capture the full heritability) and $h^2 E(r)^2 \approx 0$ we retrieve the original formula (S2).

As we discussed in the main text, several questions arise about which choices should be made when applying the expected accuracy formulas such as (S2) and (S3). Here we compare the performance of

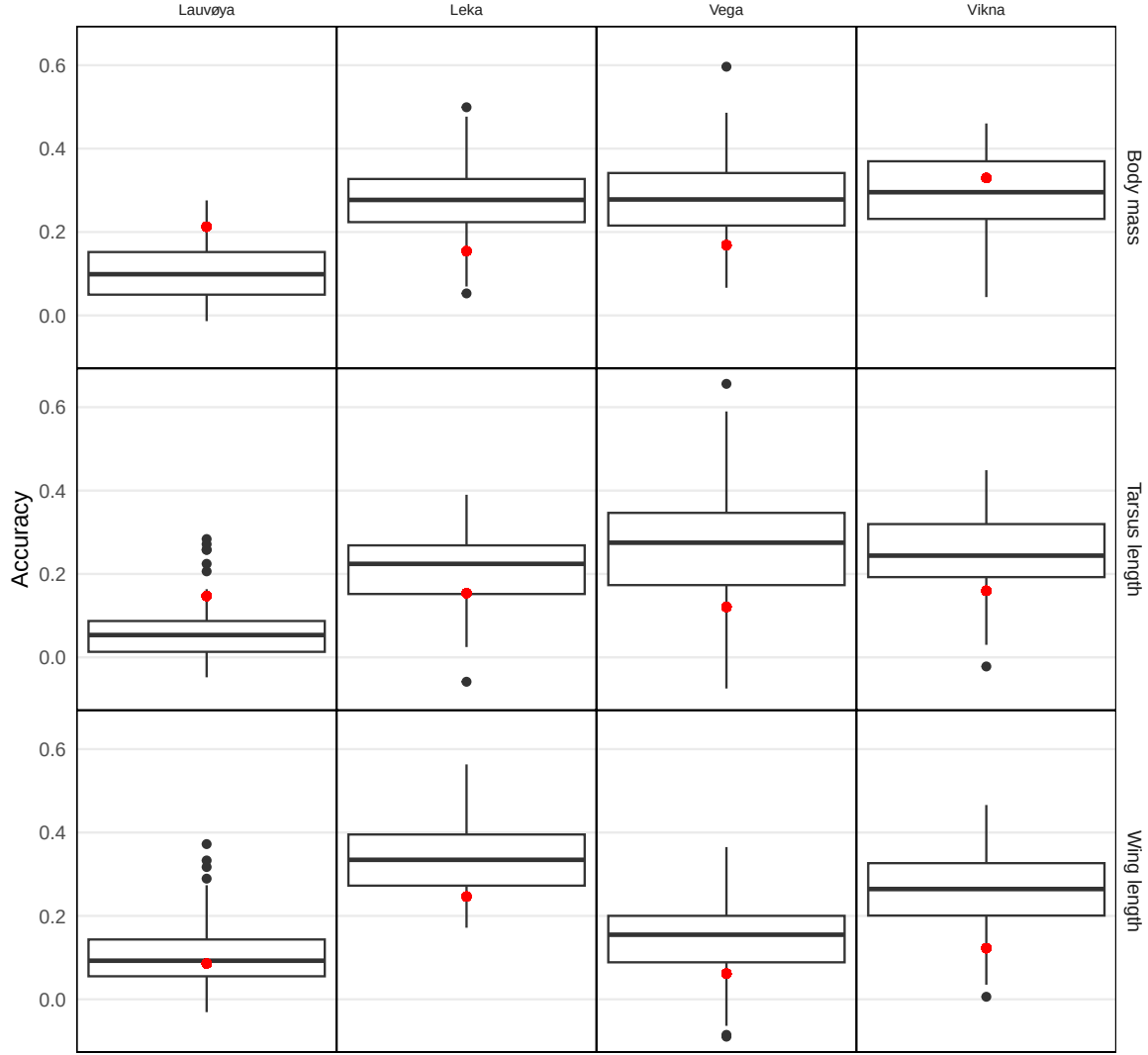


Figure S2: Accuracy in the across-population GP models for Scenario 2. Columns correspond to test sets, and rows correspond to phenotypes. The training sets are the sparrows from the southern islands, excluding the test island. The red dots are the point estimates of accuracy obtained from the original GP model and presented in the main text, while the box plots are made from the accuracies from the refitted models.

the expected accuracy formulas under combinations of choices related to M_e , heritability h^2 and which formula to use. The set of choices we considered are detailed below.

Choice of M_e estimator

We investigated 7 different estimators of M_e : the five estimators denoted M_{e1} through M_{e5} by Brard & Ricard (2015):

$$M_{e1} = \frac{2N_e L}{\log(4N_e L)},$$

$$M_{e2} = \frac{2N_e L}{\log\left(2N_e \frac{L}{N_{\text{chr}}}\right)},$$

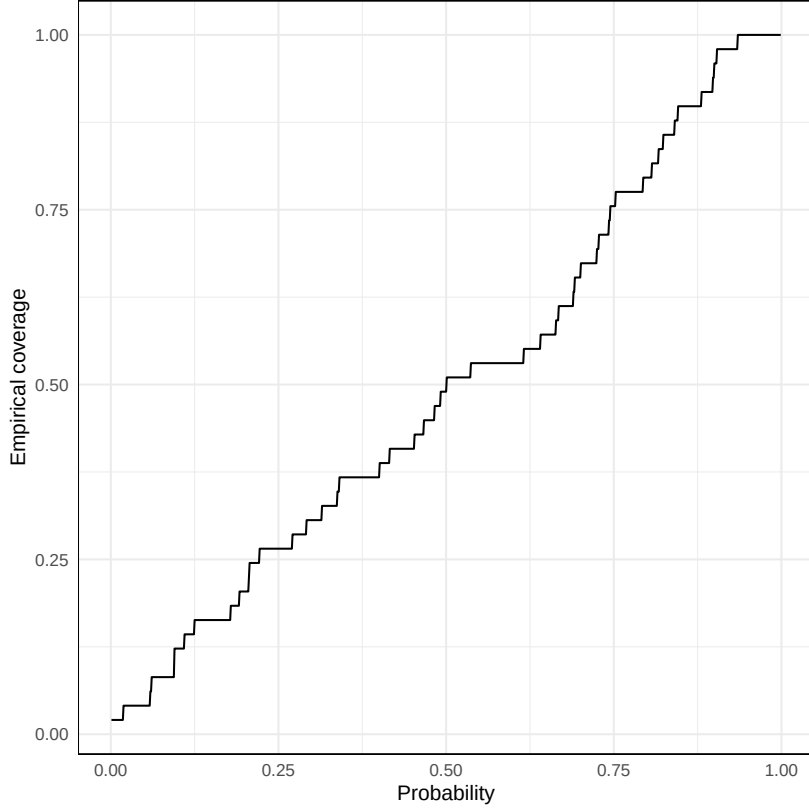


Figure S3: Empirical coverage of the resampling method. For each probability level (x -axis) we compute the proportion of models (y -axis) whose true accuracies $\hat{r}$ fall within the central π -quantile range of their corresponding resampled accuracies $\hat{r}^{(b)}$.

$$M_{e3} = \frac{2N_e L}{\log\left(N_e \frac{L}{N_c}\right)},$$

$$M_{e4} = 2N_e L,$$

$$M_{e5} = 4N_e L,$$

as well as two estimators which Lee *et al.* (2017) denote as equation (2) and (5), that we respectively label M_{e6} and M_{e7} :

$$M_{e6} = \frac{N_{\text{chr}}}{\frac{\ln\left(2N_e \frac{L}{N_{\text{chr}}} + 1\right) + 2N_e \frac{L}{N_{\text{chr}}} \left(\ln\left(2N_e \frac{L}{N_{\text{chr}}} + 1\right) - 1\right)}{4N_e^2 \frac{L^2}{N_{\text{chr}}^2}} + \frac{1}{3N_e} (N_{\text{chr}} - 1)},$$

$$M_{e7} = \tilde{G}_{\text{ac}} = \frac{1}{\widehat{\text{Var}}(G_{ii', \text{ac}})},$$

97 where $\tilde{G}_{\text{ac}}$ is as defined in the main text. Estimators M_{e1} through M_{e6} are based on the effective
 98 population size N_e , genome length L in Morgans and the number of chromosomes N_{chr} . These estima-
 99 tors use N_e , a population parameter, which thus implicitly also defines M_e as a population parameter.

Conversely, the last estimator M_{e7} can be considered as taking a model-specific view of M_e , as it considers the genomic relationships between the individuals in the training and test sets for the model.

For each model in our study we found 7 estimates of M_e by using the estimators M_{e1} through M_{e7} . To estimate N_e we used the software `currentNe` (Santiago *et al.*, 2024), whereas the parameters related to the house sparrow genome parameters were taken from Hagen *et al.* (2020). We always estimated N_e for the individuals in the *combined* training and test sets, including when considering the across-population models.

Heritability-related choices

We also investigated assumptions related to how heritability should be defined when applying expected accuracy formulas. We looked at the following heritability-related choices when using h^2 in the expected accuracy formulas (S2) and (S3):

1. Should we assume marker saturation, *i.e.*, should we assume that $h_M^2 = h^2$ or not? When assuming marker saturation we set $h_M^2 = h^2 = \hat{h}^2$. When *not* assuming marker saturation we set $h_M^2 = \hat{h}^2$ and estimated h^2 in each model using the formula $\frac{h_M^2}{h^2} = \frac{M}{M+M_e}$ (Goddard *et al.*, 2011), where M is the number of SNP markers and M_e is one of the 7 estimates outlined in the previous subsection.
2. Which subset of data should be used to estimate h^2 ? That is, should we use the heritability among the individuals in the combined training and test set, or just the individual in the test set? These heritabilities are assumed to be equal when expected accuracy formulas are derived (*e.g.*, Wientjes *et al.*, 2015). However, this question is relevant in the context of across-population GP, where the heritability of the training and test sets could be notably different. It would be problematic if the formula only performs well when using the heritability in the test set, since we would need the phenotypes in the test set to estimate this heritability, which, as explained in the main text, defeats the purpose of a formula for expected accuracy. Thus, when applying equation (S1) from Appendix S1 to estimate heritabilities, we computed $\hat{h}^2$ using samples $g_i^{(s)}$ and $\tilde{y}_{ijk}^{(s)}$ either from all individuals or just the subset of individuals in the test set.

Quadratic formula

Finally, we investigated whether one should include the term $(1 - h^2 E(r)^2)$ in the denominator of the formula, as in (S3), or not, as in (S2). That is, should one solve a quadratic equation for $E(r)^2$, or assume that $h^2 E(r)^2 \approx 0$?

Genetic correlations

(Wientjes *et al.*, 2015) suggest that in across-population GP, formulas for expected accuracy should be scaled by estimated genetic correlations between the training and test populations. We investigated whether this scaling improved the performance of the expected accuracy formulas for our across-population models, by using estimates of genetic correlation from Appendix S4.

Analysis of formula-choices

We now investigate how well the expected accuracy formulas perform depending on the choices outlined above. As we considered 7 options for M_e , and three additional binary choices (marker saturation, choice of subset for h^2 , quadratic formula or not), we investigated $7 \cdot 2 \cdot 2 \cdot 2 = 56$ combinations in the within-population case. In the across-population case we also checked whether scaling by genetic correlations improved fits, adding another binary choice, and thus increasing the number of combinations to $56 \cdot 2 = 112$. For each combination $c \in \{1, \dots, 56\}$ in the within-population case and $c \in \{1, \dots, 112\}$ in the across-population case, we evaluated how well the respective expected accuracy formula performed by computing the RMSE $\delta_c^{\mathcal{M}}$ between realized and expected accuracies;

$$\delta_c^{\mathcal{M}} = \sqrt{\frac{1}{|\mathcal{M}|} \sum_{\mathbf{m} \in \mathcal{M}} (r_{\mathbf{m}} - \mathbf{E}(r)_{\mathbf{m},c})^2},$$

where $\mathcal{M}$ is a set of models, $r_{\mathbf{m}}$ is the realized accuracy in model $\mathbf{m}$ and $\mathbf{E}(r)_{\mathbf{m},c}$ is the expected accuracy for model $\mathbf{m}$ when using the choices in combination c . We computed $\delta_c^{\mathcal{M}_{\text{within}}}$ where $\mathcal{M}_{\text{within}}$ is the set of within-population models (with $|\mathcal{M}_{\text{within}}| = 90$), and $\delta_c^{\mathcal{M}_{\text{across}}}$ where $\mathcal{M}_{\text{across}}$ is the set of across-population models (with $|\mathcal{M}_{\text{across}}| = 129$).

Table S1: Effect sizes, standard errors and associated p -values from the linear models investigating how different choices relating to expected accuracy formulas impact the expected accuracies' RMSE with regards to realized accuracy in the within-population case.

	Estimate	Std. Error	p -value
Intercept	0.1294	0.0014	< 0.0001
$X_{M_{e1}}$	0.0180	0.0016	< 0.0001
$X_{M_{e2}}$	0.0129	0.0016	< 0.0001
$X_{M_{e3}}$	0.0192	0.0016	< 0.0001
$X_{M_{e4}}$	0.3217	0.0016	< 0.0001
$X_{M_{e5}}$	0.3784	0.0016	< 0.0001
$X_{M_{e6}}$	0.0014	0.0016	0.3789
$X_{h^2=h_M^2}$	0.0014	0.0009	0.1242
$X_{h_{\text{test}}^2}$	0.0010	0.0009	0.2381
$X_{r^2 h^2 \neq 0}$	-0.0035	0.0009	0.0002

We fitted two linear regression models with $\delta_c^{\mathcal{M}_{\text{within}}}$ and $\delta_c^{\mathcal{M}_{\text{across}}}$ as the responses, respectively, and dummy variables $X_{\bullet} \in \{0, 1\}$ encoding for the different combinations as covariates. $X_{M_{e1}}$ through

Table S2: Effect sizes, standard errors and associated p -values from the linear models investigating how different choices relating to expected accuracy formulas impact the expected accuracies' RMSE with regards to realized accuracy in the across-population case.

	Estimate	Std. Error	p-value
Intercept	0.2015	0.0107	< 0.0001
$X_{M_{e1}}$	0.0423	0.0121	0.0007
$X_{M_{e2}}$	0.0450	0.0121	0.0003
$X_{M_{e3}}$	0.0418	0.0121	0.0008
$X_{M_{e4}}$	0.0539	0.0121	< 0.0001
$X_{M_{e5}}$	0.0767	0.0121	< 0.0001
$X_{M_{e6}}$	0.0535	0.0121	< 0.0001
$X_{h^2=h_M^2}$	-0.0006	0.0065	0.9230
$X_{h_{\text{test}}^2}$	-0.0007	0.0065	0.9167
$X_{r^2h^2 \neq 0}$	0.0023	0.0065	0.7283
X_{gc}	-0.0168	0.0065	0.0106

$X_{M_{e6}}$ encode for the respective choice of M_e estimator (with M_{e7} being the reference category), $X_{h^2=h_M^2} = 1$ when marker saturation ($h^2 = h_M^2$) was assumed, $X_{h_{\text{test}}^2} = 1$ when the heritability in the test was used and $X_{r^2h^2 \neq 0} = 1$ for combinations where we assume $r^2h^2 \neq 0$. $X_{\text{gc}} = 1$ when the formula for expected accuracy was scaled by estimated genetic correlations. We can thus investigate which choices (conditional on the other choices) are associated to a lower RMSE $\delta_c^{\mathcal{M}}$, *i.e.*, a better fit between the realized and expected accuracies. The results from the linear models, as shown in Tables S1 and S2, are that:

- We found very strong evidence ($p < 0.0010$) that M_{e7} performed better (*i.e.*, was associated with lower RMSE) than M_{e1} through M_{e5} , for both within-population and across-population models. For within-population models there was no evidence ($p = 0.3789$) that M_{e6} and M_{e7} performed differently, but there was very strong evidence ($p < 0.0001$) that M_{e7} performed better than M_{e6} for across-population models. Lee *et al.* (2017) states that M_{e6} and M_{e7} are equivalent, but interestingly this appears to only hold in the within-population case - in the across-population case M_{e6} is among the worst but M_{e7} is the best.
- M_{e4} and M_{e5} are drastically worse than M_{e7} in the within-case, with the overall largest predicted increase in RMSE. However, the increase in RMSE is much smaller for these estimators in the across-case, indicating that what might be a reasonable choice in one scenario could be disastrous in another.
- There was no evidence ($p > 0.1$) that assuming marker saturation ($h^2 = h_M^2$) increased the RMSE either within or across populations.
- There was no evidence ($p > 0.2$) that using the heritability in the test set degraded performance of the expected accuracy formula compared to using the heritability in the combined training

and test sets.

- We found strong evidence ($p = 0.0002$) that the quadratic version of the expected accuracy formula (*i.e.*, including the factor $(1 - E(r)^2 h^2)$ in the denominator) improved performance for within-population models, but no evidence ($p = 0.7283$) for the same in across-population models.
- We find moderate evidence ($p = 0.0106$) that scaling by genetic correlations reduces RMSE slightly in the across-population case.
- We also note that both models have an $R^2 > 0.99$, implying a lack of uncaptured interaction effects between the different choices.

The best-fitting expected accuracies for the within-population models and the across-population models are shown in Figure S4, that is, the expected accuracies obtained using the combination c with the lowest $\delta_c^{\mathcal{M}_{\text{within}}}$ and $\delta_c^{\mathcal{M}_{\text{across}}}$, respectively.¹ Evidently, the expected accuracy formula (S2) can perform well for within-population GP. However, even in the best case scenario, the formulas fail to predict the accuracy of the across-population models, usually by overestimating the accuracy. Note that Figure S4 does not necessarily imply that the formula is useful in the within-population case. The figure contains a post-hoc best fit, so the only implication is that it is *possible* to find some parameters that make formula (S2) work *given that one knows the realized accuracies*. To demonstrate how impactful the above listed choices can be, Figure S4 also contains the worst-fitting expected accuracies.² Here, the expected accuracies completely misses the realized accuracies, both in models predicting within and across populations. So, in conclusion, even under reasonable choices the formulas can perform poorly, and we find no combination of choices that makes the formula perform well for across-population GP.

Showing that $R^2 = h^2 r^2$ only if $y_i = g_i + \varepsilon_i$

Derivations of expected accuracy formulas frequently rely on the assumption that $R^2 = r^2 h^2$, where R is the phenotypic prediction accuracy $\text{Corr}(y_i, \hat{y}_i)$, r is the GP accuracy $\text{Corr}(g_i, \hat{g}_i)$ and $h^2 = \frac{\text{Var}(g_i)}{\text{Var}(y_i)}$ is the heritability. Here we show that the assumption $R^2 = r^2 h^2$ only holds in the case where there are no complicating environmental effects in the model.

¹The best set of choices for the within-population models was using M_{e7} , with $X_{h^2=h_M^2} = 0$, $X_{h^2_{\text{test}}} = 0$ and $X_{r^2 h^2 \neq 0} = 0$. The best set of choices for the across-population models was using M_{e7} , with $X_{h^2=h_M^2} = 1$, $X_{h^2_{\text{test}}} = 1$, $X_{r^2 h^2 \neq 0} = 0$ and $X_{\text{gc}} = 0$.

²The worst set of choices for the within-population models was using M_{e5} , with $X_{h^2=h_M^2} = 1$, $X_{h^2_{\text{test}}} = 0$ and $X_{r^2 h^2 \neq 0} = 0$. The worst set of choices for the across-population models was using M_{e6} , with $X_{h^2=h_M^2} = 0$, $X_{h^2_{\text{test}}} = 0$, $X_{r^2 h^2 \neq 0} = 1$ and $X_{\text{gc}} = 0$.

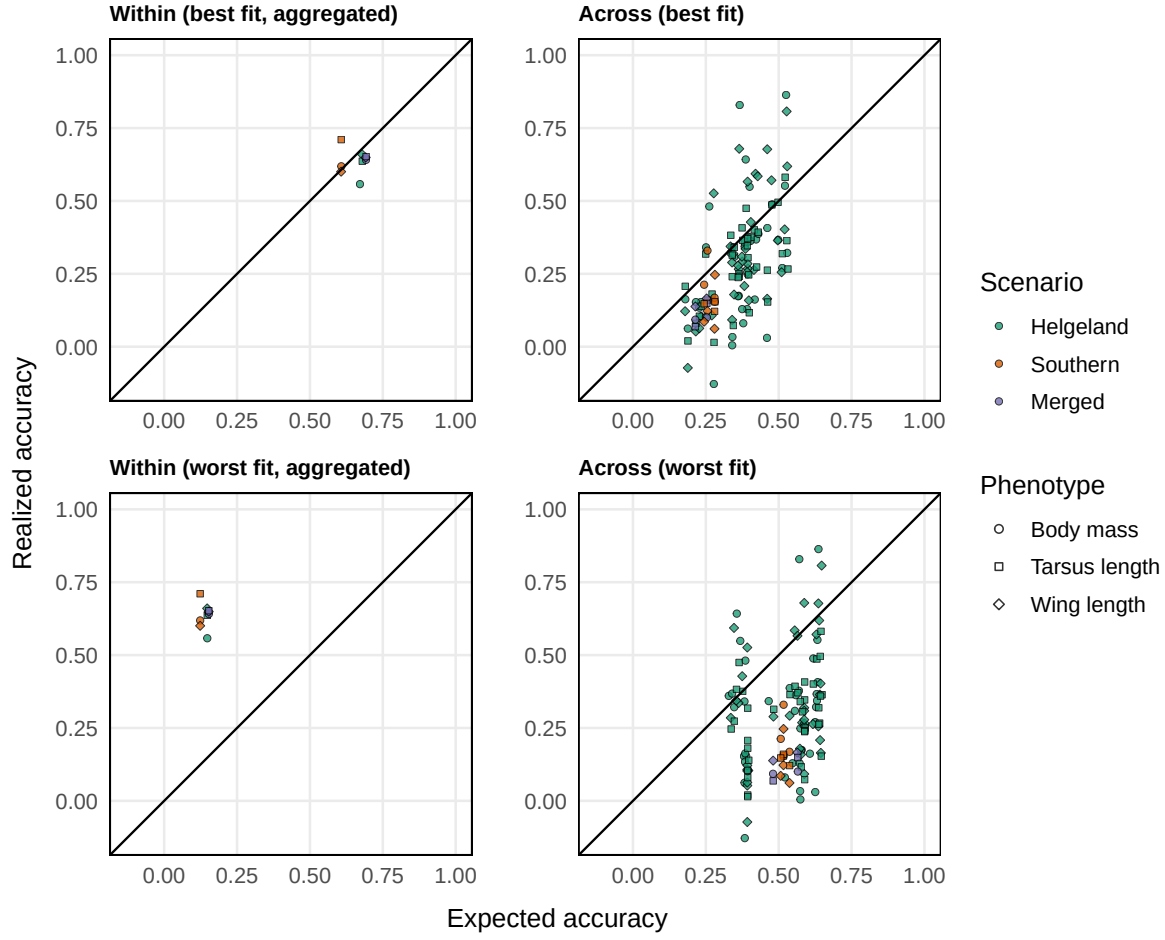


Figure S4: Realized genomic prediction accuracy vs. the expected accuracy given by a version of the expected accuracy formula. The x-axis in the top panels use the expected accuracy formula that gives the best fit (lowest RMSE) to the realized accuracies from the within-population models (left) and the across-population models (right), while the bottom panel uses the parameters that give the worst fit (highest RMSE). In the left panels, the each point corresponds to (expected or realized) accuracies averaged over the 10 folds in a given model, while in the right panels, the points corresponds to (expected or realized) accuracies in single models. Diagonals are shown as black lines, so a perfect fit would follow these lines.

We first consider the simple case where $R^2 = r^2 h^2$ does hold. Following the notation from the main text, we let

$$y_i = g_i + \epsilon_i,$$

i.e., the trait is determined only by the genetic value and a homogeneous environmental effect. Then

$$\hat{y}_i = \hat{g}_i,$$

since the environment tells us nothing about the trait. Then

$$\begin{aligned}
R^2 = \text{Corr}(y_i, \hat{y}_i) &= \frac{\text{Cov}(g_i + \epsilon_i, \hat{g}_i)}{\sqrt{\text{Var}(y_i) \text{Var}(\hat{g}_i)}} \\
&= \frac{\text{Cov}(g_i, \hat{g}_i)}{\sqrt{\text{Var}(y_i) \text{Var}(\hat{g}_i)}} \\
&= \frac{\text{Cov}(g_i, \hat{g}_i)}{\sqrt{\text{Var}(g_i) \text{Var}(\hat{g}_i)}} \cdot \sqrt{\frac{\text{Var}(g_i)}{\text{Var}(y_i)}} \\
&= \text{Corr}(g_i, \hat{g}_i) \cdot h = r \cdot h \\
\Rightarrow R^2 &= r^2 h^2.
\end{aligned}$$

Conversely, if

$$y_i = x_i + g_i + \epsilon_i,$$

i.e., i in addition to its genetic value g_i the trait value of i is determined by other (fixed or random) effects, which we lump into a single term x_i , then

$$\hat{y}_i = \hat{x}_i + \hat{g}_i.$$

In this case,

$$\begin{aligned}
\text{Corr}(y_i, \hat{y}_i) &= \frac{\text{Cov}(x_i + g_i + \epsilon_i, \hat{x}_i + \hat{g}_i)}{\sqrt{\text{Var}(y_i) \text{Var}(\hat{y}_i)}} \\
&= \frac{\text{Cov}(x_i, \hat{x}_i) + \text{Cov}(g_i, \hat{g}_i)}{\sqrt{\text{Var}(y_i) \text{Var}(\hat{y}_i)}} \\
&= \left(\frac{\text{Cov}(x_i, \hat{x}_i)}{\sqrt{\text{Var}(y_i) \text{Var}(\hat{y}_i)}} \cdot h \right) + \left(\text{Corr}(g_i, \hat{g}_i) \cdot h \cdot \frac{\text{Var}(\hat{g}_i)}{\text{Var}(\hat{y}_i)} \right) \\
&= a \cdot h + r \cdot h \cdot b \\
\Rightarrow R^2 &= (a + r \cdot h \cdot b)^2,
\end{aligned}$$

where $a = \frac{\text{Cov}(x_i, \hat{x}_i)}{\sqrt{\text{Var}(y_i) \text{Var}(\hat{y}_i)}}$ and $b = \frac{\text{Var}(\hat{g}_i)}{\text{Var}(\hat{y}_i)}$. Therefore, we will only have $R^2 = r^2 h^2$ if $a = 0$ and $b = 1$.

For $a = 0$ we must have, $\text{Cov}(x_i, \hat{x}_i) = 0$, so this situation once again reduces to a trivial case where we have no information about x_i . To conclude, we will only have $R^2 = h^2 r^2$ in a situation where

$$y_i = g_i + \epsilon_i.$$

S4: Additional population-differentiation measures

Genetic correlations

In the main text we assessed the impact of population-differences in relatedness, LD, and fixation indices on the across-population GP accuracy. Another potential factor in determining said accuracy is population-differences in the causal effects of the SNP markers (Wientjes *et al.*, 2015; Wang *et al.*, 2020). Such differences can be quantified with trait-specific genetic correlations between the populations, which measure whether the focal trait has the same genetic basis in the two populations.

To find the genetic correlations between the populations, we treated the measurements of a given trait from the two populations as two different traits. We estimated the genetic correlations using bivariate genomic animal models with response $\left(y_{ijk}^{(1)}, y_{ijk}^{(2)}\right)^\top$, where the superscript (1) denotes a measurement in the training set and (2) a measurement in the test set. The bivariate animal models included all the same effects as (1), which we modeled jointly in the two populations, except g_i and ε_{ijk} which we replaced by bivariate effects $\left(g_i^{(1)}, g_i^{(2)}\right)^\top$ and $\left(\varepsilon_{ijk}^{(1)}, \varepsilon_{ijk}^{(2)}\right)^\top$. For these bivariate effects we estimated correlations between the populations, respectively genetic correlations ρ_g and residual (environmental) correlations ρ_ε . The bivariate models were fitted in a Bayesian framework using R-INLA, using the reparameterization method given by Mathew *et al.* (2016). In their notation, we used $\text{Inv-Gamma}(\alpha = 5; \beta = 1.2)$ priors on $\sigma_{a_1}^2$ and $\sigma_{e_1}^2$, $\text{Inv-Gamma}(\alpha = 5; \beta = 0.2)$ priors on $\sigma_{a_2}^2$ and $\sigma_{e_2}^2$, and a $\text{N}(1, 0.3^2)$ prior on κ_{12} . The remaining details of the models are similar to the univariate models which we described in Appendix S1.

These bivariate models come with some caveats. Firstly, for the effects which were not treated as bivariate, it would have been preferable to estimate them independently in the two populations, rather than jointly. However, the former approach leads to a large amount of hyperparameters in R-INLA, and thus led to non-convergence for the bivariate models in Scenario 3 (Merged). For the models in Scenario 1 and 2, the results were similar with the two approaches (not shown), so we present the results from the models with joint effects, which always converged. Secondly, a drawback of the reparameterization by Mathew *et al.* (2016) is that the treatment of the two populations is not symmetric. That is, flipping which population is treated as (1) or (2) would result in a different result for the genetic correlations (which is apparent in the results for Scenario 3, see below). Thirdly, as pointed out in the main text, fitting these bivariate models relies on phenotypic measurements from both the training and test populations, so the genetic correlations are not relevant in providing *a priori* estimates of across-population GP accuracy: the phenotypes in the test set could be used to train a GP model, so this would not be in an across-population GP scenario. And, since the genetic correlations are specific to a given trait, estimates from previous across-population GP models with different traits

cannot be used to measure the genetic correlation in a new study.

Figure S5 shows the across-population accuracies plotted against the estimated genetic correlations, similarly to Figure 4 in the main text. There seems to be a slightly negative relationship between the accuracy and genetic correlations. This result is contrary to the theoretical expectation, which is that the accuracy would be higher for higher genetic correlations (Wientjes *et al.*, 2015). This seems to imply that the reason that across-population accuracy suffers is *not* due to a difference in genetic bases for the traits between populations, but again we note that the genetic correlation models should be taken with a grain of salt due to the aforementioned issues. Figure S5 also shows that the residual environmental correlations have little impact on the across-population accuracy.

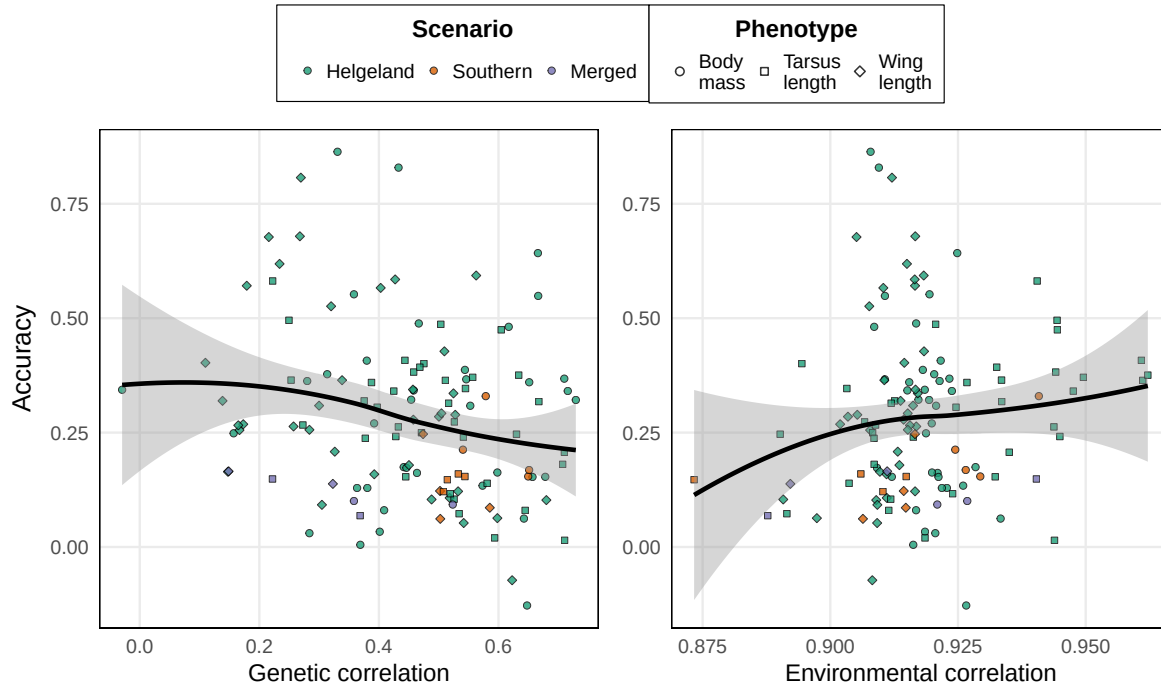


Figure S5: Scatter plot of across-population prediction accuracies vs estimated trait-specific genetic correlations and residual environmental correlations.

Heritability ratios

The full genomic animal models from S1 were used to derive heritabilities h_{AM}^2 in the test set that could be compared to the test set heritabilities h_{GP}^2 derived from the corresponding GP model. The ratio h_{GP}^2/h_{AM}^2 then indicates to what degree the GP model captured the heritability in the test set (where phenotypes per definition are unknown), compared to the situation where the test set individuals also had known phenotypes (*i.e.*, the full genomic animal models in S1). As seen in Figure S6, the ratio h_{GP}^2/h_{AM}^2 seemed to relate to across-population GP accuracy. The best-performing models had a value of h_{GP}^2/h_{AM}^2 close to 1, while we got low-to-moderate accuracies when the heritability was

underestimated ($h_{GP}^2/h_{AM}^2 < 1$), and moderate accuracies when the heritability was overestimated
 ($h_{GP}^2/h_{AM}^2 > 1$).

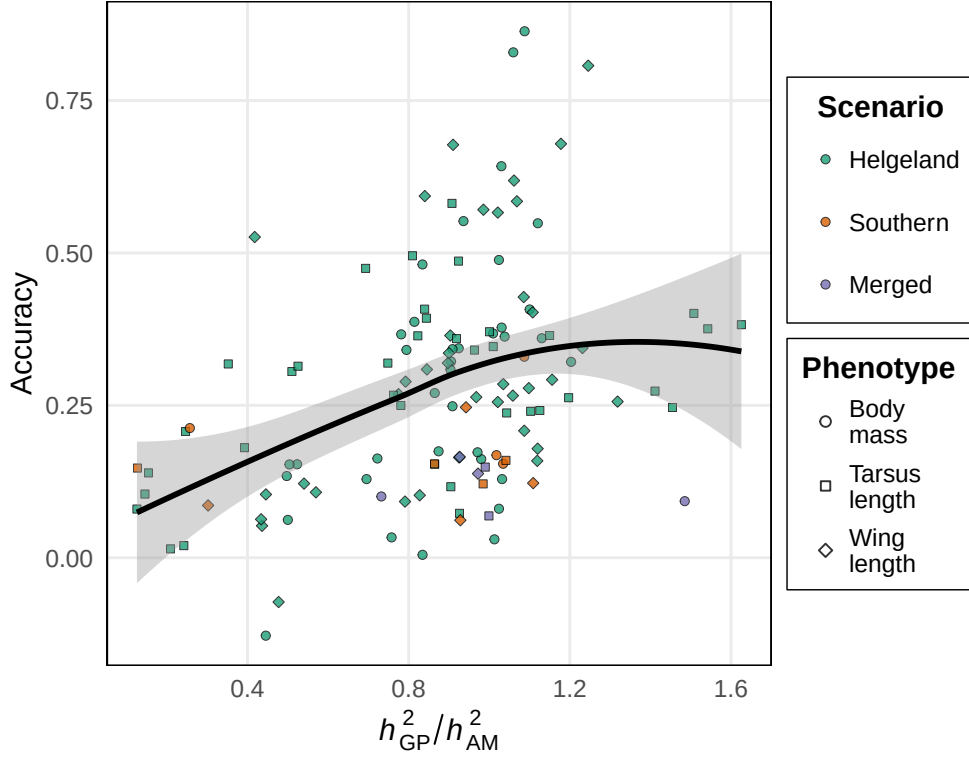


Figure S6: Scatter plot of across-population prediction accuracies vs the ratio h_{GP}^2/h_{AM}^2 between the test set heritability captured by the GP models and an animal models with access to the test set phenotypes.

S5: Variable importance for across-population accuracy

In the main text we investigated how various parameters impacted the across-population genomic prediction accuracy, namely the number of individuals in the training set N , $\bar{G}_{ac}$, $\tilde{G}_{ac}$, δLD , ΔLD , and F_{st} , while in Appendix S4 we also investigated the impact of genetic correlations ρ_g and heritability ratios h_{GP}^2/h_{AM}^2 . Figure S7 shows a pairs plot between all these aforementioned parameters, expected accuracy $E(r)$ and realized accuracy r , with scatter plots in the lower triangular part, pairwise sample correlations on the upper diagonal and histograms of each variable on the diagonal. In this calculation, we used best-performing estimator for $E(r)$ in the analysis in Appendix S3. Beyond what was discussed in the main text, we can see that F_{st} and δLD are highly correlated, while ΔLD and $E(r)$ are highly negative correlated, with a higher absolute correlation than either of the two parameters have with realized accuracies r . Thus, δLD seems to mostly reflect allele-frequency differences, while ΔLD contains much of the same information as is encoded in $E(r)$.

To see how the various parameters impact the accuracy, we fitted several multiple regression models with the across-population accuracy as response. We included four different sets of covariates in different models, and for each set we fit both a linear model (LM), as well as a generalized additive model (GAM) to capture potential non-linearity in the effects of the covariates. In *Model 1*, we included as covariates the parameters used in the main text, namely N , $\bar{G}_{ac}$, $\tilde{G}_{ac}$, δLD , ΔLD , and F_{st} . In *Model 2* we included the same covariates, in addition to parameter depending on phenotype data, namely genetic correlation ρ_g , heritability in the test set h_{AM}^2 (derived from the full animal models in Appendix S1), and the heritability ratio h_{GP}^2/h_{AM}^2 . *Model 3* contained only the expected accuracy $E(r)$, again using the best performing choices from Appendix S3, and was only fitted as a linear model since $E(r)$ should directly predict r . *Model 4* contained all previously mentioned covariates, including $E(r)$, and excluding the parameters used to compute $E(r)$, namely N , h_{AM}^2 and $\tilde{G}_{ac}$. The GAM for *Model 4* imposed a linear effect for $E(r)$ for the same reason that we did not fit a GAM for *Model 3*. We used hierarchical partitioning, as implemented in the R packages `g1mm.hp` (Lai *et al.*, 2022) and `gam.hp` (Lai *et al.*, 2024), to partition the variance and deviance explained by the LMs and GAMs, respectively, into parts attributable to each covariate.

Table S3 contains the results from the variable importance analysis, that is, from the hierarchical partitioning of variance (LMs) or deviance (GAMs) explained by Models 1 through 4. We found that $\bar{G}_{ac}$, $\tilde{G}_{ac}$, and ΔLD were the most important variables in all LM where they were included. δLD and F_{st} were only given importance in the non-linear models. Notably, the LMs never explain more than 40% of the variance, and even when using GAMs, we at best only explain about 55% of the deviance (Model 4). The additional information from consider the parameters relying on phenotypes was marginal, as found by comparing models 1 and 2. Comparing Models 1 and 3, we found that our parameters together explain more variance in the across-population than the formula for expected accuracy does.

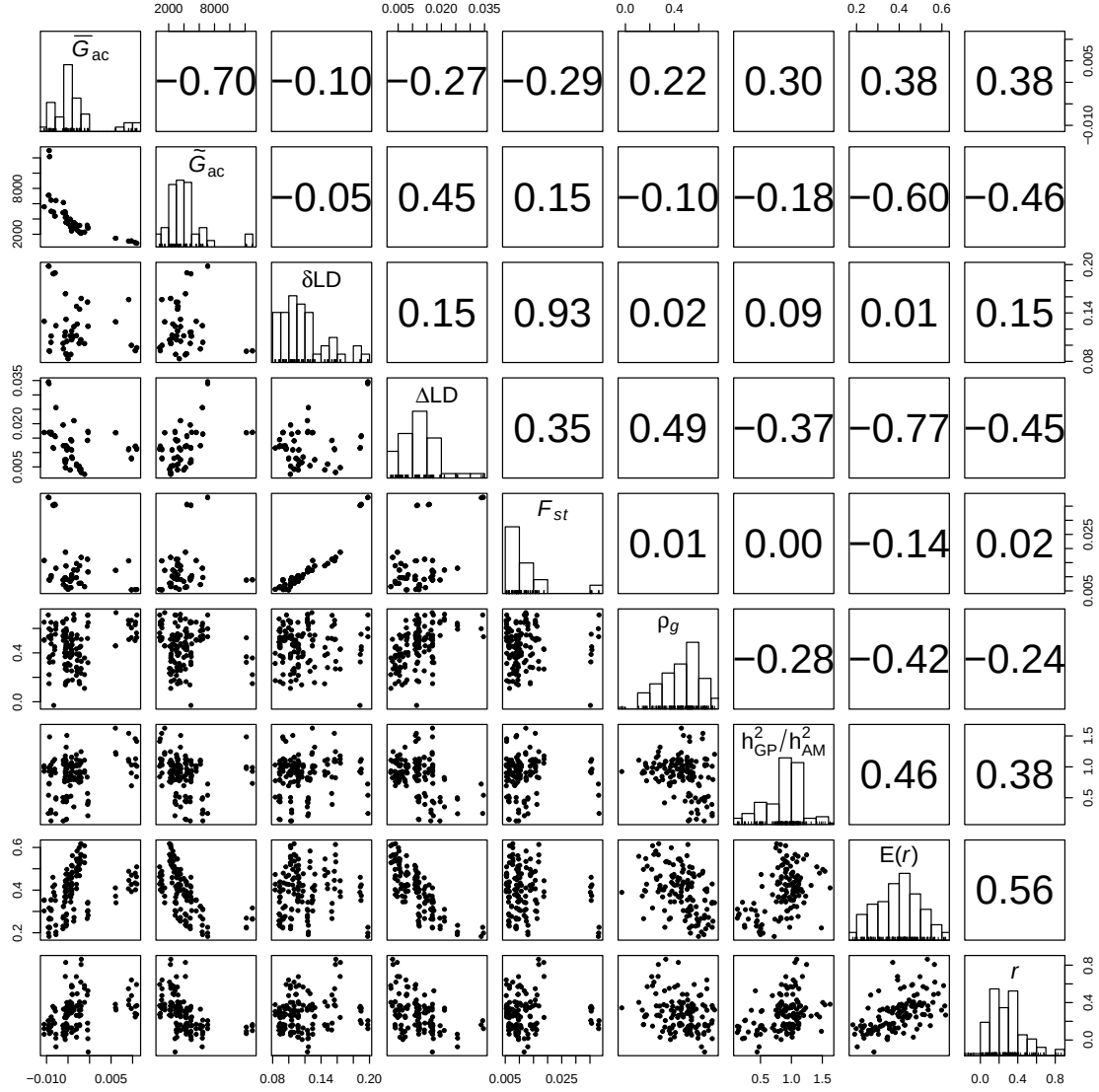


Figure S7: Pairwise scatter plots (lower triangular matrix) and sample Pearson correlations (upper triangular matrix) between population-differentiation measures ($\bar{G}_{ac}$, $\tilde{G}_{ac}$, δLD , ΔLD , F_{st} , genetic correlation ρ_g , heritability ratio h_{GP}^2/h_{AM}^2), expected accuracy $E(r)$ and realized accuracy r in the across-population models. The diagonals show histograms for each variable.

Table S3: Results from a variable importance analysis of the impact of various parameters upon across-population GP prediction accuracy. Each row shows how much variance or deviance is explained by a given covariate in a given model for the LMs or GAMs, respectively, according to hierarchical partitioning. The percentages in parentheses indicates what proportion of the total variance/deviance explained by that model is attributed to the given covariate. A dash indicates the covariate was not included in the given model. The bottom row indicates how much variance/deviance was explained by the LMs/GAMs, respectively.

	Model 1		Model 2		Model 3	Model 4	
	Linear	GAM	Linear	GAM	Linear	Linear	GAM
N	0.0582 (15.54%)	0.0634 (12.12%)	0.0330 (8.42%)	0.0423 (7.77%)	-	-	-
$\bar{G}_{ac}$	0.0884 (23.61%)	0.0638 (12.20%)	0.0751 (19.16%)	0.0595 (10.93%)	-	0.0803 (21.37%)	0.0815 (14.78%)
$\tilde{G}_{ac}$	0.0979 (26.15%)	0.0987 (18.87%)	0.0972 (24.80%)	0.0910 (16.72%)	-	-	-
δLD	0.0238 (6.36%)	0.1086 (20.76%)	0.0217 (5.54%)	0.1050 (19.29%)	-	0.0259 (6.89%)	0.1244 (22.56%)
ΔLD	0.0915 (24.44%)	0.1009 (19.29%)	0.0707 (18.04%)	0.0859 (15.78%)	-	0.0735 (19.56%)	0.1010 (18.32%)
F_{st}	0.0146 (3.90%)	0.0876 (16.75%)	0.0127 (3.24%)	0.0884 (16.24%)	-	0.0146 (3.89%)	0.1072 (19.44%)
ρ_g	-	-	0.0268 (6.84%)	0.0211 (3.88%)	-	0.0256 (6.81%)	0.0307 (5.57%)
h_{AM}^2	-	-	0.0013 (0.33%)	0.0092 (1.69%)	-	-	-
h_{GP}^2/h_{AM}^2	-	-	0.0535 (13.65%)	0.0419 (7.70%)	-	0.0524 (13.94%)	0.0495 (8.98%)
$E(r)$	-	-	-	-	0.2620 (100%)	0.1035 (27.54%)	0.0570 (10.34%)
Total	0.3744 (100%)	0.5230 (100%)	0.3921 (100%)	0.5443 (100%)	0.2620 (100%)	0.3758 (100%)	0.5513 (100%)

S6: Supplemental tables

Genomic animal models

Body mass

Table S4: Statistics related to the genomic animal models with body mass as response. For statistics derived from the model, we show posterior means and the 0.025 and 0.975 quantiles of the posterior distribution in parenthesis. N is the number of individuals, N_{obs} is the number of observations, M is the number of SNPs used, d is the value that was added to the diagonal of the GRM for the respective model, μ is the intercept, $\beta_{\text{sex=m}}$ is the effect of being male compared to female, $\beta_{\text{month}X}$ is the effect of month X compared to January, β_{age} is the effect of age, σ_G^2 is the additive genetic variance, $\sigma_{\text{hatch-year}}^2$ is the variance explained by hatch year, σ_{island}^2 is the variance explained by island of measurement, σ_{id}^2 is the variance explained by permanent environmental effects, $\sigma_{\text{session}}^2$ is the variance explained by measurement session and σ_ϵ^2 is the residual variance.

Scenario	Helgeland	Southern	Merged
N	3456	2254	5710
N_{obs}	7977	4124	12101
M	65221	64917	64943
d	6.6e-03	5.0e-03	6.2e-03
μ	32.22 (30.96, 33.47)	30.51 (29.83, 31.19)	31.41 (30.74, 32.09)
$\beta_{\text{sex=m}}$	-0.32 (-0.44, -0.20)	0.41 (0.28, 0.54)	-0.06 (-0.16, 0.03)
$\beta_{\text{month}=2}$	0.12 (-1.27, 1.52)	0.11 (-0.53, 0.75)	-0.09 (-0.75, 0.57)
$\beta_{\text{month}=3}$	-0.33 (-1.98, 1.33)	-0.35 (-1.00, 0.30)	-0.57 (-1.24, 0.10)
$\beta_{\text{month}=4}$	-0.39 (-1.72, 0.93)	-0.25 (-1.03, 0.54)	-0.13 (-0.87, 0.61)
$\beta_{\text{month}=5}$	0.10 (-1.14, 1.34)	1.34 (0.41, 2.27)	0.63 (-0.05, 1.30)
$\beta_{\text{month}=6}$	-0.28 (-1.52, 0.96)	1.22 (0.32, 2.13)	0.26 (-0.42, 0.93)
$\beta_{\text{month}=7}$	-0.80 (-2.04, 0.43)	-0.57 (-1.42, 0.29)	-0.29 (-0.96, 0.39)
$\beta_{\text{month}=8}$	-0.37 (-1.61, 0.87)	-0.09 (-1.25, 1.07)	0.16 (-0.52, 0.83)
$\beta_{\text{month}=9}$	0.23 (-1.02, 1.49)	1.62 (0.76, 2.48)	0.81 (0.12, 1.51)
$\beta_{\text{month}=10}$	0.10 (-1.14, 1.34)	1.51 (0.69, 2.33)	0.63 (-0.05, 1.31)
$\beta_{\text{month}=11}$	0.12 (-1.18, 1.41)	0.89 (0.03, 1.76)	0.63 (-0.10, 1.37)
β_{age}	0.05 (0.01, 0.09)	0.09 (0.05, 0.13)	0.05 (0.02, 0.08)
σ_G^2	1.30 (1.07, 1.58)	1.42 (1.18, 1.70)	1.24 (1.07, 1.43)
$\sigma_{\text{hatch-year}}^2$	0.09 (0.04, 0.18)	0.07 (0.01, 0.24)	0.06 (0.02, 0.12)
σ_{island}^2	0.07 (0.01, 0.21)	0.02 (0.00, 0.10)	0.05 (0.01, 0.15)
σ_{id}^2	0.71 (0.53, 0.92)	0.35 (0.20, 0.56)	0.66 (0.53, 0.80)
$\sigma_{\text{session}}^2$	2.72 (2.59, 2.85)	1.32 (1.21, 1.44)	2.30 (2.21, 2.40)
σ_ϵ^2	0.21 (0.19, 0.23)	0.28 (0.24, 0.32)	0.23 (0.21, 0.25)

Table S5: Statistics related to the genomic animal models with tarsus length as response. For statistics derived from the model, we show posterior means and the 0.025 and 0.975 quantiles of the posterior distribution in parenthesis. N is the number of individuals, N_{obs} is the number of observations, M is the number of SNPs used, d is the value that was added to the diagonal of the GRM for the respective model, μ is the intercept, $\beta_{\text{sex=m}}$ is the effect of being male compared to female, $\beta_{\text{month}X}$ is the effect of month X compared to January, β_{age} is the effect of age, σ_G^2 is the additive genetic variance, $\sigma_{\text{hatch-year}}^2$ is the variance explained by hatch year, σ_{island}^2 is the variance explained by island of measurement, σ_{id}^2 is the variance explained by permanent environmental effects, $\sigma_{\text{session}}^2$ is the variance explained by measurement session and σ_ε^2 is the residual variance.

Scenario	Helgeland	Southern	Merged
N	3445	2255	5700
N_{obs}	8067	4171	12238
M	65221	64917	64943
d	6.6e-03	5.0e-03	6.1e-03
μ	19.23 (19.07, 19.38)	19.46 (19.32, 19.61)	19.35 (19.26, 19.45)
$\beta_{\text{sex=m}}$	0.08 (0.03, 0.13)	0.11 (0.05, 0.17)	0.09 (0.06, 0.13)
$\beta_{\text{month}=2}$	0.01 (-0.31, 0.34)	0.03 (-0.07, 0.13)	0.07 (-0.01, 0.16)
$\beta_{\text{month}=3}$	0.32 (0.14, 0.51)	-0.02 (-0.12, 0.08)	0.02 (-0.06, 0.11)
$\beta_{\text{month}=4}$	0.12 (-0.03, 0.28)	-0.03 (-0.17, 0.10)	0.05 (-0.05, 0.14)
$\beta_{\text{month}=5}$	0.14 (-0.00, 0.28)	0.19 (0.05, 0.33)	0.08 (-0.01, 0.16)
$\beta_{\text{month}=6}$	0.19 (0.05, 0.34)	0.25 (0.11, 0.39)	0.13 (0.04, 0.21)
$\beta_{\text{month}=7}$	0.21 (0.07, 0.36)	0.18 (0.05, 0.31)	0.15 (0.06, 0.23)
$\beta_{\text{month}=8}$	0.22 (0.08, 0.36)	0.09 (-0.08, 0.26)	0.15 (0.06, 0.23)
$\beta_{\text{month}=9}$	0.32 (0.18, 0.46)	-0.01 (-0.14, 0.12)	0.23 (0.15, 0.32)
$\beta_{\text{month}=10}$	0.26 (0.12, 0.40)	0.13 (0.00, 0.25)	0.19 (0.11, 0.28)
$\beta_{\text{month}=11}$	0.19 (0.04, 0.34)	0.09 (-0.04, 0.22)	0.13 (0.04, 0.22)
β_{age}	0.00 (-0.00, 0.01)	-0.02 (-0.02, -0.01)	-0.01 (-0.01, -0.00)
σ_G^2	0.25 (0.21, 0.29)	0.24 (0.20, 0.30)	0.24 (0.21, 0.27)
$\sigma_{\text{hatch-year}}^2$	0.01 (0.00, 0.02)	0.02 (0.00, 0.06)	0.01 (0.00, 0.02)
σ_{island}^2	0.00 (0.00, 0.01)	0.00 (0.00, 0.02)	0.00 (0.00, 0.01)
σ_{id}^2	0.35 (0.33, 0.39)	0.34 (0.30, 0.38)	0.35 (0.33, 0.38)
$\sigma_{\text{session}}^2$	0.02 (0.02, 0.02)	0.02 (0.01, 0.02)	0.02 (0.02, 0.02)
σ_ε^2	0.01 (0.01, 0.01)	0.02 (0.01, 0.02)	0.01 (0.01, 0.02)

Table S6: Statistics related to the genomic animal models with wing length as response. For statistics derived from the model, we show posterior means and the 0.025 and 0.975 quantiles of the posterior distribution in parenthesis. N is the number of individuals, N_{obs} is the number of observations, M is the number of SNPs used, d is the value that was added to the diagonal of the GRM for the respective model, μ is the intercept, $\beta_{\text{sex=m}}$ is the effect of being male compared to female, $\beta_{\text{month}X}$ is the effect of month X compared to January, β_{age} is the effect of age, σ_G^2 is the additive genetic variance, $\sigma_{\text{hatch-year}}^2$ is the variance explained by hatch year, σ_{island}^2 is the variance explained by island of measurement, σ_{id}^2 is the variance explained by permanent environmental effects, $\sigma_{\text{session}}^2$ is the variance explained by measurement session and σ_ε^2 is the residual variance.

Scenario	Helgeland	Southern	Merged
N	3424	2254	5678
N_{obs}	7997	4151	12148
M	65220	64917	64943
d	6.6e-03	5.0e-03	6.1e-03
μ	78.00 (77.08, 78.91)	77.66 (77.07, 78.25)	77.63 (77.10, 78.16)
$\beta_{\text{sex=m}}$	2.73 (2.64, 2.83)	2.67 (2.55, 2.78)	2.71 (2.63, 2.78)
$\beta_{\text{month}=2}$	0.77 (-0.94, 2.48)	0.78 (0.24, 1.31)	0.64 (0.15, 1.13)
$\beta_{\text{month}=3}$	-0.20 (-1.34, 0.94)	0.71 (0.16, 1.25)	0.53 (0.03, 1.04)
$\beta_{\text{month}=4}$	-0.33 (-1.28, 0.61)	0.87 (0.21, 1.54)	0.28 (-0.28, 0.83)
$\beta_{\text{month}=5}$	0.07 (-0.81, 0.95)	0.71 (-0.06, 1.47)	0.41 (-0.09, 0.92)
$\beta_{\text{month}=6}$	-0.19 (-1.08, 0.69)	1.07 (0.31, 1.84)	0.16 (-0.34, 0.66)
$\beta_{\text{month}=7}$	-0.42 (-1.30, 0.47)	-0.75 (-1.46, -0.04)	-0.08 (-0.59, 0.42)
$\beta_{\text{month}=8}$	-0.35 (-1.23, 0.54)	-1.12 (-2.04, -0.19)	-0.01 (-0.52, 0.49)
$\beta_{\text{month}=9}$	-1.11 (-2.01, -0.22)	0.10 (-0.70, 0.90)	-0.72 (-1.24, -0.20)
$\beta_{\text{month}=10}$	0.25 (-0.64, 1.13)	0.16 (-0.52, 0.84)	0.58 (0.07, 1.08)
$\beta_{\text{month}=11}$	0.65 (-0.27, 1.57)	0.74 (0.02, 1.46)	0.95 (0.40, 1.50)
β_{age}	0.33 (0.30, 0.35)	0.26 (0.22, 0.29)	0.31 (0.29, 0.33)
σ_G^2	1.50 (1.32, 1.72)	1.26 (1.06, 1.50)	1.32 (1.18, 1.47)
$\sigma_{\text{hatch-year}}^2$	0.05 (0.02, 0.12)	0.12 (0.04, 0.28)	0.11 (0.05, 0.22)
σ_{island}^2	0.14 (0.04, 0.37)	0.03 (0.00, 0.19)	0.09 (0.03, 0.21)
σ_{id}^2	0.30 (0.20, 0.42)	0.25 (0.13, 0.41)	0.33 (0.25, 0.43)
$\sigma_{\text{session}}^2$	0.92 (0.84, 0.99)	0.70 (0.61, 0.79)	0.86 (0.80, 0.92)
σ_ε^2	0.49 (0.45, 0.54)	0.36 (0.31, 0.42)	0.45 (0.42, 0.49)

Table S7: Statistics related to each within-population GP model with body mass as response. N is the number of individuals in the training set, N_{test} is the number of individuals in the test set, N_{obs} is the number of observations in the training set, M is the number of SNPs used, $\hat{r}$ is the estimated GP accuracy, $\hat{\lambda}$ is the scaling factor used in finding $\hat{r}$ as described in the main text, $\bar{G}_{\text{ac}}$ is the mean relatedness between the training and test set individuals $\tilde{G}_{\text{ac}}$ is the precision of the relatedness between training and test set individuals and d is the value that was added to the diagonal of the GRM for the respective model.

Scenario	N	N_{test}	N_{obs}	M	$\hat{r}$	$\hat{\lambda}$	$\bar{G}_{\text{ac}}$	$\tilde{G}_{\text{ac}}$	d
Helgeland	3110	346	7189	65221	0.68	0.51	-3.0E-04	994.44	6.6E-03
Helgeland	3110	346	7183	65221	0.51	0.46	-2.9E-04	1025.52	6.6E-03
Helgeland	3111	345	7183	65221	0.52	0.52	-3.1E-04	967.35	6.6E-03
Helgeland	3110	346	7118	65221	0.58	0.52	-2.9E-04	969.79	6.6E-03
Helgeland	3111	345	7193	65221	0.63	0.49	-3.0E-04	973.11	6.6E-03
Helgeland	3110	346	7148	65221	0.57	0.51	-3.0E-04	973.66	6.6E-03
Helgeland	3111	345	7219	65221	0.55	0.49	-2.9E-04	927.85	6.6E-03
Helgeland	3110	346	7166	65221	0.38	0.55	-3.2E-04	1049.53	6.6E-03
Helgeland	3111	345	7154	65221	0.46	0.54	-3.0E-04	985.07	6.6E-03
Helgeland	3110	346	7240	65221	0.68	0.50	-2.8E-04	977.08	6.6E-03
Southern	2028	226	3743	64917	0.47	0.65	-4.6E-04	957.85	5.0E-03
Southern	2029	225	3705	64917	0.62	0.60	-4.4E-04	949.37	5.0E-03
Southern	2029	225	3677	64917	0.52	0.64	-4.4E-04	960.83	5.0E-03
Southern	2028	226	3705	64917	0.72	0.63	-4.7E-04	940.40	5.0E-03
Southern	2029	225	3724	64917	0.50	0.67	-4.5E-04	912.83	5.0E-03
Southern	2029	225	3725	64917	0.69	0.61	-4.3E-04	928.00	5.0E-03
Southern	2028	226	3695	64917	0.79	0.63	-4.5E-04	922.91	5.0E-03
Southern	2029	225	3745	64917	0.68	0.59	-4.6E-04	974.15	5.0E-03
Southern	2029	225	3677	64917	0.61	0.64	-4.5E-04	914.58	5.0E-03
Southern	2028	226	3720	64917	0.59	0.66	-4.4E-04	946.65	5.0E-03
Merged	5139	571	10911	64943	0.66	0.51	-2.0E-04	1499.93	6.2E-03
Merged	5139	571	10914	64943	0.58	0.47	-1.7E-04	1489.37	6.2E-03
Merged	5139	571	10898	64943	0.66	0.54	-1.8E-04	1499.87	6.2E-03
Merged	5139	571	10910	64943	0.68	0.50	-1.8E-04	1470.63	6.2E-03
Merged	5139	571	10890	64943	0.68	0.51	-1.9E-04	1584.31	6.2E-03
Merged	5139	571	10881	64943	0.58	0.50	-1.8E-04	1543.50	6.2E-03
Merged	5139	571	10950	64943	0.62	0.54	-1.8E-04	1529.31	6.2E-03
Merged	5139	571	10803	64943	0.64	0.51	-1.8E-04	1512.03	6.2E-03
Merged	5139	571	10862	64943	0.57	0.55	-1.9E-04	1463.24	6.2E-03
Merged	5139	571	10890	64943	0.73	0.52	-1.8E-04	1546.83	6.2E-03

Within-population GP: Tarsus length

Table S8: Statistics related to each within-population GP model with tarsus length as response. N is the number of individuals in the training set, N_{test} is the number of individuals in the test set, N_{obs} is the number of observations in the training set, M is the number of SNPs used, $\hat{r}$ is the estimated GP accuracy, $\hat{\lambda}$ is the scaling factor used in finding $\hat{r}$ as described in the main text, $\bar{G}_{\text{ac}}$ is the mean relatedness between the training and test set individuals $\tilde{G}_{\text{ac}}$ is the precision of the relatedness between training and test set individuals and d is the value that was added to the diagonal of the GRM for the respective model.

Scenario	N	N_{test}	N_{obs}	M	$\hat{r}$	$\hat{\lambda}$	$\bar{G}_{\text{ac}}$	$\tilde{G}_{\text{ac}}$	d
Helgeland	3100	345	7213	65221	0.71	0.65	-2.8E-04	952.73	6.6E-03
Helgeland	3101	344	7276	65221	0.62	0.67	-3.0E-04	966.56	6.6E-03
Helgeland	3100	345	7282	65221	0.57	0.57	-3.0E-04	1000.38	6.6E-03
Helgeland	3101	344	7271	65221	0.71	0.56	-2.8E-04	950.44	6.6E-03
Helgeland	3100	345	7247	65221	0.64	0.64	-3.1E-04	941.31	6.6E-03
Helgeland	3101	344	7285	65221	0.63	0.66	-3.0E-04	989.68	6.6E-03
Helgeland	3101	344	7251	65221	0.56	0.68	-3.1E-04	995.56	6.6E-03
Helgeland	3100	345	7258	65221	0.61	0.59	-2.9E-04	1014.26	6.6E-03
Helgeland	3101	344	7286	65221	0.71	0.63	-2.9E-04	975.53	6.6E-03
Helgeland	3100	345	7234	65221	0.59	0.67	-3.2E-04	1010.86	6.6E-03
Southern	2029	226	3777	64917	0.61	0.65	-4.3E-04	992.45	5.0E-03
Southern	2030	225	3771	64917	0.67	0.64	-4.4E-04	932.69	5.0E-03
Southern	2029	226	3729	64917	0.83	0.61	-4.6E-04	954.08	5.0E-03
Southern	2030	225	3763	64917	0.89	0.60	-4.4E-04	867.55	5.0E-03
Southern	2029	226	3777	64917	0.52	0.72	-4.4E-04	936.92	5.0E-03
Southern	2030	225	3765	64917	0.74	0.66	-4.6E-04	936.71	5.0E-03
Southern	2030	225	3742	64917	0.68	0.69	-4.4E-04	944.22	5.0E-03
Southern	2029	226	3740	64917	0.59	0.72	-4.7E-04	941.27	5.0E-03
Southern	2030	225	3746	64917	0.77	0.62	-4.9E-04	948.05	5.0E-03
Southern	2029	226	3729	64917	0.80	0.66	-4.2E-04	965.73	5.0E-03
Merged	5130	570	11004	64943	0.56	0.64	-1.9E-04	1505.09	6.1E-03
Merged	5130	570	11013	64943	0.71	0.66	-1.8E-04	1478.44	6.1E-03
Merged	5130	570	10996	64943	0.58	0.64	-1.9E-04	1521.87	6.1E-03
Merged	5130	570	10996	64943	0.72	0.62	-1.8E-04	1492.34	6.1E-03
Merged	5130	570	11019	64943	0.70	0.60	-1.8E-04	1516.68	6.1E-03
Merged	5130	570	11017	64943	0.61	0.66	-1.8E-04	1517.12	6.1E-03
Merged	5130	570	11029	64943	0.70	0.68	-1.8E-04	1500.58	6.1E-03
Merged	5130	570	11003	64943	0.70	0.64	-1.8E-04	1534.69	6.1E-03
Merged	5130	570	11021	64943	0.56	0.64	-1.8E-04	1537.99	6.1E-03
Merged	5130	570	11044	64943	0.69	0.65	-1.8E-04	1486.97	6.1E-03

Within-population GP: Wing length

Table S9: Statistics related to each within-population GP model with wing length as response. N is the number of individuals in the training set, N_{test} is the number of individuals in the test set, N_{obs} is the number of observations in the training set, M is the number of SNPs used, $\hat{r}$ is the estimated GP accuracy, $\hat{\lambda}$ is the scaling factor used in finding $\hat{r}$ as described in the main text, $\bar{G}_{\text{ac}}$ is the mean relatedness between the training and test set individuals $\tilde{G}_{\text{ac}}$ is the precision of the relatedness between training and test set individuals and d is the value that was added to the diagonal of the GRM for the respective model.

Scenario	N	N_{test}	N_{obs}	M	$\hat{r}$	$\hat{\lambda}$	$\bar{G}_{\text{ac}}$	$\tilde{G}_{\text{ac}}$	d
Helgeland	3081	343	7179	65220	0.56	0.56	-3.0E-04	1011.74	6.6E-03
Helgeland	3082	342	7150	65220	0.45	0.54	-2.9E-04	986.15	6.6E-03
Helgeland	3082	342	7217	65220	0.74	0.52	-3.2E-04	1028.35	6.6E-03
Helgeland	3081	343	7248	65220	0.93	0.51	-3.2E-04	939.75	6.6E-03
Helgeland	3082	342	7222	65220	0.53	0.56	-3.1E-04	981.62	6.6E-03
Helgeland	3082	342	7187	65220	0.52	0.53	-3.0E-04	940.36	6.6E-03
Helgeland	3081	343	7202	65220	0.50	0.55	-3.1E-04	991.61	6.6E-03
Helgeland	3082	342	7199	65220	0.62	0.58	-2.9E-04	973.20	6.6E-03
Helgeland	3082	342	7198	65220	0.75	0.55	-3.1E-04	978.97	6.6E-03
Helgeland	3081	343	7171	65220	0.98	0.47	-3.1E-04	963.80	6.6E-03
Southern	2028	226	3755	64917	0.52	0.51	-4.6E-04	959.22	5.0E-03
Southern	2029	225	3738	64917	0.57	0.52	-5.0E-04	913.87	5.0E-03
Southern	2029	225	3753	64917	0.19	0.60	-4.4E-04	936.63	5.0E-03
Southern	2028	226	3735	64917	0.59	0.54	-4.5E-04	955.92	5.0E-03
Southern	2029	225	3752	64917	0.67	0.56	-4.8E-04	955.69	5.0E-03
Southern	2029	225	3706	64917	0.60	0.56	-4.7E-04	939.30	5.0E-03
Southern	2028	226	3764	64917	0.66	0.51	-4.8E-04	952.82	5.0E-03
Southern	2029	225	3725	64917	0.86	0.49	-4.7E-04	983.14	5.0E-03
Southern	2029	225	3686	64917	0.62	0.55	-4.6E-04	882.99	5.0E-03
Southern	2028	226	3745	64917	0.72	0.49	-4.5E-04	955.63	5.0E-03
Merged	5110	568	10874	64943	0.76	0.52	-1.9E-04	1517.27	6.1E-03
Merged	5110	568	10942	64943	0.43	0.53	-1.9E-04	1511.77	6.1E-03
Merged	5110	568	10893	64943	0.70	0.50	-2.0E-04	1516.07	6.1E-03
Merged	5111	567	10941	64943	0.56	0.53	-1.9E-04	1522.85	6.1E-03
Merged	5110	568	10938	64943	0.63	0.51	-1.9E-04	1527.88	6.1E-03
Merged	5110	568	10992	64943	0.63	0.52	-1.9E-04	1565.06	6.1E-03
Merged	5111	567	10929	64943	0.66	0.50	-1.8E-04	1459.52	6.1E-03
Merged	5110	568	10951	64943	0.58	0.52	-1.8E-04	1472.10	6.1E-03
Merged	5110	568	10969	64943	0.67	0.50	-1.8E-04	1540.39	6.1E-03
Merged	5110	568	10903	64943	0.86	0.49	-1.8E-04	1512.90	6.1E-03

Table S10: Statistics related to each across-population GP model with body mass as response. N is the number of individuals in the training set, N_{test} is the number of individuals in the test set, N_{obs} is the number of observations in the training set, M is the number of SNPs used, $\hat{r}$ is the estimated GP accuracy, $\hat{\lambda}$ is the scaling factor used in finding $\hat{r}$ as described in the main text, $\frac{h_{\text{GP}}^2}{h_{\text{AM}}^2}$ is the heritability ratio described in the main text, $\bar{G}_{\text{ac}}$ is the mean relatedness between the training and test set individuals $\tilde{G}_{\text{ac}}$ is the precision of the relatedness between training and test set individuals, ΔLD1 is LD measure 1, ΔLD2 is LD measure 2, F_{st} is the population-pairwise fixation index between the training and test set, ρ_g is the trait-specific estimated genetic correlation between the training and test set, ρ_ε is the estimated residual (environmental) correlation between the training and test set and d is the value that was added to the diagonal of the GRM for the respective model.

Train islands	Test islands	N	N_{test}	N_{obs}	M	$\hat{r}$	$\hat{\lambda}$	$\frac{h_{\text{GP}}^2}{h_{\text{AM}}^2}$	$\bar{G}_{\text{ac}}$	$\tilde{G}_{\text{ac}}$	ΔLD1	ΔLD2	F_{st}	ρ_g	ρ_ε	d
Helgeland	Nesøy	3323	133	7544	65221	0.41	0.50	1.10	-0.002	3173.80	0.0015	0.052	0.014	0.38	0.92	6.6E-03
Helgeland	Myken	3368	88	7837	65221	0.86	0.50	1.09	-0.002	2278.34	0.0012	0.057	0.017	0.33	0.91	6.6E-03
Helgeland	Træna	3184	272	7616	65221	0.27	0.54	0.86	-0.002	2307.47	0.0021	0.037	0.008	0.39	0.92	6.6E-03
Helgeland	Selvær	3234	222	7689	65221	0.32	0.55	0.91	-0.002	2144.63	0.0017	0.039	0.008	0.45	0.92	6.6E-03
Helgeland	Gjerøy	2913	543	6534	65221	0.16	0.50	0.98	-0.005	3550.86	0.0043	0.038	0.009	0.46	0.92	6.6E-03
Helgeland	Hestmannøy	2423	1033	5203	65221	0.38	0.50	1.03	-0.005	3380.29	0.0045	0.030	0.006	0.31	0.92	6.6E-03
Helgeland	Indre Kvarøy	3085	371	7135	65221	0.49	0.49	1.02	-0.004	2692.72	0.0031	0.040	0.010	0.47	0.92	6.6E-03
Helgeland	Lurøy-Onøy	3190	266	7362	65221	0.03	0.52	1.01	-0.003	3068.63	0.0022	0.040	0.010	0.28	0.92	6.6E-03
Helgeland	Lovund	3309	147	7796	65221	0.55	0.52	0.94	-0.001	2284.41	0.0010	0.036	0.006	0.36	0.92	6.6E-03
Helgeland	Sleneset	3267	189	7742	65221	0.37	0.57	0.78	-0.003	2547.18	0.0021	0.046	0.012	0.55	0.91	6.6E-03
Helgeland	Aldra	3283	173	7354	65221	0.34	0.55	0.92	-0.008	4908.19	0.0042	0.069	0.035	-0.03	0.92	6.6E-03
Helgeland	Non-farm	2538	918	6772	65221	0.17	0.53	0.87	-0.005	4315.31	0.0048	0.033	0.006	0.44	0.93	6.6E-03
Helgeland	Farm	937	2519	1247	65221	0.15	0.50	0.52	-0.005	4196.60	0.0048	0.032	0.006	0.68	0.91	6.6E-03
Non-farm	Nesøy	918	133	1205	65221	0.48	0.46	0.83	-0.001	3212.31	0.0054	0.055	0.016	0.62	0.91	1.0E-09
Non-farm	Myken	830	88	1065	65217	0.55	0.51	1.12	0.009	1122.54	0.0042	0.056	0.016	0.67	0.91	1.0E-09
Non-farm	Træna	646	272	844	65217	0.36	0.55	1.13	0.011	897.66	0.0049	0.034	0.005	0.65	0.92	1.0E-09
Non-farm	Selvær	696	222	917	65217	0.37	0.53	1.01	0.011	826.21	0.0046	0.035	0.006	0.71	0.92	1.0E-09
Non-farm	Gjerøy	918	543	1205	65220	0.06	0.51	0.50	-0.008	6435.98	0.0099	0.045	0.013	0.64	0.93	1.0E-09
Non-farm	Hestmannøy	918	1033	1205	65221	0.15	0.50	0.50	-0.006	4833.89	0.0076	0.038	0.009	0.66	0.92	1.0E-09
Non-farm	Indre Kvarøy	918	371	1205	65221	0.34	0.49	0.79	-0.004	3531.31	0.0082	0.045	0.012	0.72	0.92	1.0E-09
Non-farm	Lurøy-Onøy	918	266	1205	65212	-0.13	0.52	0.45	-0.000	2837.14	0.0065	0.045	0.012	0.65	0.93	1.0E-09
Non-farm	Lovund	771	147	1024	65217	0.64	0.53	1.03	0.010	1129.47	0.0030	0.035	0.005	0.67	0.92	1.0E-09
Non-farm	Sleneset	729	189	970	65217	0.32	0.58	1.20	0.006	1483.43	0.0066	0.047	0.012	0.73	0.92	1.0E-09
Non-farm	Aldra	918	173	1205	65221	0.16	0.50	0.72	-0.009	7107.91	0.0126	0.073	0.038	0.60	0.92	1.0E-09
Non-farm	Farm	918	2519	1205	65221	0.13	0.51	0.50	-0.005	4341.38	0.0049	0.033	0.006	0.57	0.93	4.0E-03
Farm	Nesøy	2386	133	6297	65221	0.36	0.49	1.04	-0.003	3169.44	0.0023	0.053	0.015	0.28	0.92	1.0E-09
Farm	Myken	2519	88	6730	65221	0.83	0.49	1.06	-0.006	4193.67	0.0018	0.059	0.019	0.43	0.91	1.0E-09
Farm	Træna	2519	272	6730	65221	0.03	0.54	0.76	-0.006	4927.24	0.0029	0.040	0.009	0.40	0.92	1.0E-09
Farm	Selvær	2519	222	6730	65221	0.00	0.54	0.83	-0.006	4954.24	0.0026	0.042	0.010	0.37	0.92	1.0E-09
Farm	Gjerøy	1976	543	5287	65221	0.08	0.51	1.02	-0.004	3047.01	0.0055	0.039	0.009	0.41	0.92	1.0E-09
Farm	Hestmannøy	1486	1033	3956	65221	0.34	0.50	0.91	-0.004	2909.69	0.0056	0.032	0.006	0.46	0.92	1.0E-09
Farm	Indre Kvarøy	2148	371	5888	65221	0.39	0.50	0.81	-0.004	2437.54	0.0043	0.041	0.010	0.54	0.92	1.0E-09
Farm	Lurøy-Onøy	2253	266	6115	65221	0.13	0.50	1.03	-0.004	3202.95	0.0033	0.043	0.011	0.38	0.92	1.0E-09
Farm	Lovund	2519	147	6730	65221	0.13	0.53	0.70	-0.004	3977.23	0.0016	0.040	0.008	0.36	0.92	1.0E-09
Farm	Sleneset	2519	189	6730	65221	0.25	0.59	0.91	-0.006	3544.56	0.0029	0.048	0.014	0.16	0.92	1.0E-09
Farm	Aldra	2346	173	6107	65221	0.31	0.53	0.90	-0.008	4381.58	0.0057	0.070	0.035	0.55	0.92	1.0E-09
Farm	Non-farm	2519	918	6730	65221	0.17	0.52	0.97	-0.005	4341.38	0.0049	0.033	0.006	0.45	0.91	4.0E-03
Southern	Leka	1666	588	2947	64917	0.15	0.64	1.03	-0.009	5010.31	0.0066	0.041	0.010	0.65	0.93	5.0E-03
Southern	Vega	1845	409	3259	64917	0.17	0.60	1.02	-0.011	5598.08	0.0065	0.047	0.016	0.65	0.93	5.0E-03
Southern	Vikna	1668	586	3143	64917	0.33	0.61	1.09	-0.006	6151.58	0.0048	0.035	0.007	0.58	0.94	5.0E-03
Southern	Lauvøya	1589	665	3036	64917	0.21	0.67	0.26	-0.009	6462.24	0.0064	0.038	0.010	0.54	0.92	5.0E-03
Helgeland	Southern	3461	2249	7987	64943	0.10	0.60	0.73	-0.009	12959.11	0.0066	0.034	0.009	0.36	0.93	6.2E-03
Southern	Helgeland	2268	3442	4156	64943	0.09	0.50	1.49	-0.009	12168.65	0.0066	0.034	0.009	0.52	0.92	6.2E-03

Table S11: Statistics related to each across-population GP model with tarsus length as response. N is the number of individuals in the training set, N_{test} is the number of individuals in the test set, N_{obs} is the number of observations in the training set, M is the number of SNPs used, $\hat{r}$ is the estimated GP accuracy, $\hat{\lambda}$ is the scaling factor used in finding $\hat{r}$ as described in the main text, $\frac{h_{\text{GP}}^2}{h_{\text{AM}}^2}$ is the heritability ratio described in the main text, $\bar{G}_{\text{ac}}$ is the mean relatedness between the training and test set individuals $\tilde{G}_{\text{ac}}$ is the precision of the relatedness between training and test set individuals, ΔLD1 is LD measure 1, ΔLD2 is LD measure 2, F_{st} is the population-pairwise fixation index between the training and test set, ρ_g is the trait-specific estimated genetic correlation between the training and test set, ρ_ε is the estimated residual (environmental) correlation between the training and test set and d is the value that was added to the diagonal of the GRM for the respective model.

Train islands	Test islands	N	N_{test}	N_{obs}	M	$\hat{r}$	$\hat{\lambda}$	$\frac{h_{\text{GP}}^2}{h_{\text{AM}}^2}$	$\bar{G}_{\text{ac}}$	$\tilde{G}_{\text{ac}}$	ΔLD1	ΔLD2	F_{st}	ρ_g	ρ_ε	d
Helgeland	Nesøy	3314	131	7640	65221	0.15	0.75	0.86	-0.002	3148.54	0.0015	0.052	0.014	0.44	0.93	6.6E-03
Helgeland	Myken	3356	89	7927	65221	0.36	0.70	0.82	-0.002	2239.83	0.0012	0.057	0.017	0.51	0.96	6.6E-03
Helgeland	Træna	3176	269	7698	65221	0.32	0.71	0.75	-0.002	2295.74	0.0020	0.037	0.008	0.38	0.91	6.6E-03
Helgeland	Selvær	3223	222	7772	65221	0.27	0.74	0.76	-0.002	2108.90	0.0016	0.038	0.008	0.27	0.91	6.6E-03
Helgeland	Gjerøy	2901	544	6607	65221	0.40	0.55	1.51	-0.005	3568.50	0.0043	0.038	0.009	0.48	0.89	6.6E-03
Helgeland	Hestmannøy	2423	1022	5264	65221	0.31	0.66	0.51	-0.005	3385.84	0.0045	0.030	0.006	0.40	0.92	6.6E-03
Helgeland	Indre Kvarøy	3075	370	7188	65221	0.49	0.65	0.92	-0.004	2696.11	0.0031	0.040	0.009	0.50	0.92	6.6E-03
Helgeland	Lurøy-Onøy	3178	267	7467	65221	0.26	0.59	1.20	-0.003	3030.53	0.0022	0.040	0.010	0.43	0.94	6.6E-03
Helgeland	Lovund	3297	148	7882	65221	0.58	0.65	0.91	-0.001	2267.87	0.0009	0.037	0.007	0.22	0.94	6.6E-03
Helgeland	Sleneset	3256	189	7820	65221	0.50	0.68	0.81	-0.003	2550.02	0.0021	0.046	0.012	0.25	0.94	6.6E-03
Helgeland	Aldra	3270	175	7449	65221	0.36	0.69	0.92	-0.009	4911.51	0.0043	0.069	0.035	0.39	0.93	6.6E-03
Helgeland	Non-farm	2528	917	6831	65221	0.25	0.70	0.78	-0.005	4285.73	0.0048	0.033	0.006	0.47	0.91	6.6E-03
Helgeland	Farm	936	2509	1280	65221	0.14	0.62	0.15	-0.005	4168.77	0.0047	0.032	0.006	0.58	0.90	6.6E-03
Non-farm	Nesøy	917	131	1236	65221	0.18	0.71	0.39	-0.000	2973.36	0.0055	0.055	0.016	0.71	0.91	1.0E-09
Non-farm	Myken	828	89	1096	65217	0.38	0.73	1.54	0.009	1088.02	0.0044	0.056	0.016	0.63	0.96	1.0E-09
Non-farm	Træna	648	269	867	65217	0.25	0.76	1.45	0.011	892.76	0.0047	0.034	0.005	0.63	0.89	1.0E-09
Non-farm	Selvær	695	222	941	65217	0.27	0.75	1.41	0.011	817.33	0.0044	0.035	0.005	0.53	0.91	1.0E-09
Non-farm	Gjerøy	917	544	1236	65220	0.02	0.56	0.24	-0.008	6407.81	0.0100	0.045	0.013	0.59	0.92	1.0E-09
Non-farm	Hestmannøy	917	1022	1236	65221	0.08	0.64	0.13	-0.006	4825.20	0.0076	0.038	0.009	0.64	0.91	1.0E-09
Non-farm	Indre Kvarøy	917	370	1236	65221	0.32	0.67	0.35	-0.004	3559.23	0.0082	0.045	0.012	0.67	0.93	1.0E-09
Non-farm	Lurøy-Onøy	917	267	1236	65212	0.01	0.57	0.21	-0.000	2819.28	0.0067	0.045	0.012	0.71	0.94	1.0E-09
Non-farm	Lovund	769	148	1051	65217	0.47	0.68	0.69	0.010	1116.33	0.0031	0.035	0.005	0.60	0.94	1.0E-09
Non-farm	Sleneset	728	189	989	65217	0.38	0.69	1.63	0.006	1479.48	0.0066	0.047	0.012	0.46	0.94	1.0E-09
Non-farm	Aldra	917	175	1236	65221	0.21	0.66	0.25	-0.010	7105.12	0.0128	0.073	0.038	0.71	0.94	1.0E-09
Non-farm	Farm	917	2509	1236	65221	0.10	0.62	0.15	-0.005	4311.29	0.0048	0.033	0.006	0.53	0.91	4.0E-03
Farm	Nesøy	2378	131	6360	65221	0.12	0.72	0.90	-0.003	3233.92	0.0023	0.053	0.015	0.52	0.92	1.0E-09
Farm	Myken	2509	89	6787	65221	0.24	0.70	1.13	-0.006	4203.09	0.0019	0.059	0.019	0.43	0.94	1.0E-09
Farm	Træna	2509	269	6787	65221	0.24	0.73	1.10	-0.006	4925.65	0.0029	0.041	0.010	0.54	0.92	1.0E-09
Farm	Selvær	2509	222	6787	65221	0.07	0.74	0.93	-0.006	4803.36	0.0025	0.042	0.010	0.53	0.89	1.0E-09
Farm	Gjerøy	1965	544	5327	65221	0.36	0.57	1.15	-0.004	3066.64	0.0055	0.039	0.009	0.25	0.93	1.0E-09
Farm	Hestmannøy	1487	1022	3984	65221	0.31	0.64	0.53	-0.004	2919.50	0.0057	0.032	0.006	0.52	0.91	1.0E-09
Farm	Indre Kvarøy	2139	370	5908	65221	0.39	0.67	0.84	-0.004	2435.04	0.0042	0.041	0.010	0.47	0.93	1.0E-09
Farm	Lurøy-Onøy	2242	267	6187	65221	0.37	0.62	1.00	-0.004	3153.37	0.0033	0.043	0.011	0.56	0.95	1.0E-09
Farm	Lovund	2509	148	6787	65221	0.41	0.69	0.84	-0.005	3968.44	0.0015	0.040	0.008	0.44	0.96	1.0E-09
Farm	Sleneset	2509	189	6787	65221	0.35	0.73	1.01	-0.005	3550.59	0.0030	0.048	0.014	0.54	0.90	1.0E-09
Farm	Aldra	2334	175	6169	65221	0.34	0.69	0.96	-0.008	4383.92	0.0058	0.070	0.036	0.42	0.95	1.0E-09
Farm	Non-farm	2509	917	6787	65221	0.24	0.72	1.04	-0.005	4311.29	0.0048	0.033	0.006	0.38	0.91	4.0E-03
Southern	Leka	1667	588	2993	64917	0.15	0.70	0.86	-0.009	5008.84	0.0066	0.041	0.010	0.54	0.91	5.0E-03
Southern	Vega	1847	408	3344	64917	0.12	0.67	0.98	-0.011	5599.24	0.0065	0.047	0.016	0.51	0.91	5.0E-03
Southern	Vikna	1667	588	3182	64917	0.16	0.64	1.04	-0.006	6154.38	0.0048	0.035	0.007	0.53	0.91	5.0E-03
Southern	Lauvøya	1590	665	3007	64917	0.15	0.73	0.13	-0.009	6468.81	0.0063	0.038	0.010	0.51	0.87	5.0E-03
Helgeland	Southern	3450	2250	8077	64943	0.15	0.63	0.99	-0.009	12956.45	0.0066	0.034	0.009	0.22	0.94	6.1E-03
Southern	Helgeland	2269	3431	4205	64943	0.07	0.65	1.00	-0.009	12165.15	0.0066	0.034	0.009	0.37	0.89	6.1E-03

Table S12: Statistics related to each across-population GP model with wing length as response. N is the number of individuals in the training set, N_{test} is the number of individuals in the test set, N_{obs} is the number of observations in the training set, M is the number of SNPs used, $\hat{r}$ is the estimated GP accuracy, $\hat{\lambda}$ is the scaling factor used in finding $\hat{r}$ as described in the main text, $\frac{h_{\text{GP}}^2}{h_{\text{AM}}^2}$ is the heritability ratio described in the main text, $\bar{G}_{\text{ac}}$ is the mean relatedness between the training and test set individuals $\tilde{G}_{\text{ac}}$ is the precision of the relatedness between training and test set individuals, ΔLD1 is LD measure 1, ΔLD2 is LD measure 2, F_{st} is the population-pairwise fixation index between the training and test set, ρ_g is the trait-specific estimated genetic correlation between the training and test set, ρ_ε is the estimated residual (environmental) correlation between the training and test set and d is the value that was added to the diagonal of the GRM for the respective model.

Train islands	Test islands	N	N_{test}	N_{obs}	M	$\hat{r}$	$\hat{\lambda}$	$\frac{h_{\text{GP}}^2}{h_{\text{AM}}^2}$	$\bar{G}_{\text{ac}}$	$\tilde{G}_{\text{ac}}$	ΔLD1	ΔLD2	F_{st}	ρ_g	ρ_ε	d
Helgeland	Nesøy	3292	132	7569	65220	0.16	0.56	0.93	-0.002	3154.59	0.0015	0.052	0.014	0.15	0.91	6.6E-03
Helgeland	Myken	3335	89	7857	65220	0.81	0.47	1.25	-0.002	2233.41	0.0012	0.057	0.017	0.27	0.91	6.6E-03
Helgeland	Træna	3157	267	7634	65220	0.26	0.55	1.02	-0.002	2316.74	0.0020	0.037	0.008	0.17	0.91	6.6E-03
Helgeland	Selvær	3204	220	7705	65220	0.62	0.50	1.06	-0.002	2123.52	0.0016	0.038	0.008	0.23	0.92	6.6E-03
Helgeland	Gjerøy	2883	541	6547	65220	0.26	0.56	0.97	-0.005	3573.12	0.0043	0.038	0.009	0.26	0.92	6.6E-03
Helgeland	Hestmannøy	2409	1015	5232	65220	0.27	0.56	0.77	-0.005	3380.30	0.0046	0.030	0.006	0.17	0.90	6.6E-03
Helgeland	Indre Kvarøy	3057	367	7119	65220	0.57	0.52	0.99	-0.004	2701.24	0.0031	0.040	0.009	0.18	0.92	6.6E-03
Helgeland	Lurøy-Onøy	3158	266	7414	65220	0.68	0.54	0.91	-0.003	3021.66	0.0021	0.040	0.010	0.22	0.91	6.6E-03
Helgeland	Lovund	3276	148	7813	65220	0.40	0.54	1.11	-0.001	2274.08	0.0010	0.037	0.007	0.11	0.91	6.6E-03
Helgeland	Sleneset	3238	186	7752	65220	0.36	0.55	0.90	-0.003	2547.44	0.0022	0.046	0.012	0.34	0.91	6.6E-03
Helgeland	Aldra	3250	174	7372	65220	0.21	0.51	1.09	-0.008	4900.34	0.0043	0.069	0.035	0.33	0.91	6.6E-03
Helgeland	Non-farm	2514	910	6773	65220	0.27	0.53	1.06	-0.005	4274.74	0.0048	0.033	0.006	0.16	0.92	6.6E-03
Helgeland	Farm	929	2495	1268	65220	0.10	0.54	0.45	-0.005	4157.16	0.0047	0.032	0.006	0.49	0.89	6.6E-03
Non-farm	Nesøy	910	132	1224	65220	0.11	0.57	0.57	-0.000	2971.26	0.0054	0.055	0.016	0.52	0.91	1.0E-09
Non-farm	Myken	821	89	1084	65216	0.43	0.50	1.09	0.009	1082.63	0.0044	0.056	0.016	0.51	0.92	1.0E-09
Non-farm	Træna	643	267	861	65216	0.28	0.55	1.03	0.011	907.01	0.0048	0.034	0.006	0.50	0.90	1.0E-09
Non-farm	Selvær	690	220	932	65216	0.59	0.52	0.84	0.011	828.17	0.0044	0.035	0.006	0.56	0.92	1.0E-09
Non-farm	Gjerøy	910	541	1224	65219	-0.07	0.56	0.48	-0.008	6390.60	0.0100	0.045	0.013	0.62	0.91	1.0E-09
Non-farm	Hestmannøy	910	1015	1224	65220	0.05	0.52	0.44	-0.006	4817.53	0.0076	0.038	0.009	0.54	0.91	1.0E-09
Non-farm	Indre Kvarøy	910	367	1224	65220	0.10	0.52	0.83	-0.004	3535.06	0.0082	0.045	0.012	0.68	0.91	1.0E-09
Non-farm	Lurøy-Onøy	910	266	1224	65212	0.53	0.54	0.42	-0.000	2819.93	0.0067	0.045	0.012	0.32	0.91	1.0E-09
Non-farm	Lovund	762	148	1040	65216	0.34	0.51	0.90	0.010	1121.73	0.0032	0.036	0.005	0.52	0.92	1.0E-09
Non-farm	Sleneset	724	186	979	65216	0.34	0.60	1.23	0.006	1479.89	0.0066	0.047	0.012	0.46	0.92	1.0E-09
Non-farm	Aldra	910	174	1224	65220	0.12	0.55	0.54	-0.010	7095.88	0.0130	0.073	0.038	0.53	0.92	1.0E-09
Non-farm	Farm	910	2495	1224	65220	0.06	0.55	0.43	-0.005	4300.35	0.0048	0.033	0.006	0.60	0.90	3.9E-03
Farm	Nesøy	2363	132	6301	65220	0.16	0.54	1.12	-0.003	3243.48	0.0023	0.053	0.015	0.39	0.91	1.0E-09
Farm	Myken	2495	89	6729	65220	0.68	0.49	1.18	-0.006	4195.79	0.0018	0.059	0.019	0.27	0.92	1.0E-09
Farm	Træna	2495	267	6729	65220	0.09	0.54	0.79	-0.006	4915.69	0.0029	0.041	0.009	0.30	0.91	1.0E-09
Farm	Selvær	2495	220	6729	65220	0.32	0.52	0.90	-0.006	4777.19	0.0025	0.042	0.010	0.14	0.91	1.0E-09
Farm	Gjerøy	1954	541	5279	65220	0.29	0.54	1.16	-0.004	3074.25	0.0055	0.039	0.009	0.50	0.92	1.0E-09
Farm	Hestmannøy	1480	1015	3964	65220	0.29	0.55	0.79	-0.004	2916.43	0.0057	0.032	0.006	0.53	0.91	1.0E-09
Farm	Indre Kvarøy	2128	367	5851	65220	0.58	0.52	1.07	-0.004	2446.65	0.0043	0.041	0.010	0.43	0.92	1.0E-09
Farm	Lurøy-Onøy	2229	266	6146	65220	0.57	0.56	1.02	-0.004	3139.56	0.0033	0.043	0.011	0.40	0.91	1.0E-09
Farm	Lovund	2495	148	6729	65220	0.31	0.55	0.85	-0.005	3959.41	0.0015	0.040	0.008	0.30	0.92	1.0E-09
Farm	Sleneset	2495	186	6729	65220	0.26	0.55	1.32	-0.005	3543.74	0.0029	0.048	0.014	0.28	0.92	1.0E-09
Farm	Aldra	2321	174	6104	65220	0.18	0.49	1.12	-0.008	4373.07	0.0058	0.070	0.036	0.45	0.91	1.0E-09
Farm	Non-farm	2495	910	6729	65220	0.28	0.52	1.10	-0.005	4300.35	0.0048	0.033	0.006	0.46	0.91	3.9E-03
Southern	Leka	1668	586	2977	64917	0.25	0.56	0.94	-0.009	5007.22	0.0066	0.041	0.010	0.47	0.92	5.0E-03
Southern	Vega	1845	409	3324	64917	0.06	0.53	0.93	-0.011	5602.19	0.0065	0.047	0.016	0.50	0.91	5.0E-03
Southern	Vikna	1666	588	3161	64917	0.12	0.50	1.11	-0.006	6151.02	0.0049	0.035	0.007	0.50	0.91	5.0E-03
Southern	Lauvøya	1589	665	3004	64917	0.09	0.54	0.30	-0.009	6475.60	0.0064	0.038	0.010	0.59	0.91	5.0E-03
Helgeland	Southern	3429	2249	8007	64943	0.17	0.55	0.93	-0.009	12958.62	0.0066	0.034	0.009	0.15	0.91	6.1E-03
Southern	Helgeland	2268	3410	4185	64943	0.14	0.50	0.97	-0.009	12163.28	0.0066	0.034	0.009	0.32	0.89	6.1E-03

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